

# Curve fitting algorithm for multimodal particle size distributions – a theoretical basis

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Source code for helper read functions are housed here <https://github.com/christopher-rapp/multimodal.git>

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## 1 Setup

### 1.1 Retrieve Read Functions

### 1.2 Import Libraries

```
library(stringr)
library(data.table)
library(tidyr)
```

```
library(dplyr)
library(lubridate)
library(logr)
library(ggplot2)
library(purrr)
library(patchwork)
library(ncdf4)
```

## 1.3 User Paths

To download the example data for BMI and TSI visit. For NASA-AMES or netCDF examples see linked sources below.

<https://github.com/christopher-rapp/multimodal/tree/main/data>

Define user paths such as ‘~/Library/Documents/multimodal/example/BMI/’

## 2. Formatting Data

Raw view of data format is appended to the bottom of this document

### 2.1 Brechtel Manufacturing Inc. (BMI) Data

```
BMI.data.ls <- readPSD_BMI(import.path.BMI, tz = "US/Eastern")

# Read functions export data as a list to account for multiple files in a directory
dataPSD.BMI <- BMI.data.ls[[1]]
```

### 2.2 TSI Data

Use example data from Storm Peak Laboratory for a new particle formation event (NPF) on 2022-03-23 MDT.

```
TSI.data.ls <- readPSD_TSI(import.path.TSI, tz = "US/Mountain")

# Read functions export data as a list to account for multiple files in a directory
dataPSD.TSI <- TSI.data.ls[[1]]
```

### 2.3 netCDF Data

Data were obtained from the Atmospheric Radiation Measurement (ARM) User Facility, a U.S. Department of Energy (DOE) Office of Science user facility managed by the Biological and Environmental Research Program.

SGP SMPS data were obtained from the Atmospheric Radiation Measurement (ARM) User Facility, a U.S. Department of Energy (DOE) Office of Science user facility managed by the Biological and Environmental Research Program.

Kuang, C., Singh, A., Howie, J., Salwen, C., & Hayes, C. Scanning mobility particle sizer (AOSSMPS), 2016-11-15 to 2025-06-23, Southern Great Plains (SGP), Lamont, OK (Extended and Co-located with C1) (E13). Atmospheric Radiation Measurement (ARM) User Facility. <https://doi.org/10.5439/1476898>

Similarly for the BMI and TSI data, define the import.path of downloaded NC

```
NC.data.ls <- readPSD_NC(import.path.NC)

# Read functions export data as a list to account for multiple files in a directory
dataPSD.NC <- NC.data.ls[[1]]
```

## 2.4 NASA-AMES Data

Data obtained from <https://ebas-data.nilu.no/DataSets.aspx?stations=US9050R&InstrumentTypes=smps&fromDate=1970-01-01&toDate=2025-12-31>.

The EBAS database has largely been funded by the UN-ECE CLRTAP (EMEP), AMAP and through NILU internal resources. Specific developments have been possible due to projects like EUSAAR (EU-FP5)(EBAS web interface), EBAS-Online (Norwegian Research Council INFRA) (upgrading of database platform) and HTAP (European Commission DG-ENV)(import and export routines to build a secondary repository in support of [www.htap.org](http://www.htap.org)). A large number of specific projects have supported development of data and meta data reporting schemes in dialog with data providers (EU)(CREATE, ACTRIS and others).

```
NAS.data.ls <- readPSD_NAS(import.path.NAS)

# Read functions export data as a list to account for multiple files in a directory
dataPSD.NAS <- NAS.data.ls[[1]]
```

## 3. Running multimodal

Let's run multimodal on an example dataset using a Brechtel SEMS (Model 2002). Note the log path will need to be changed to whatever location you'd like it sent to!

### 3.1 Example 1 - Laboratory Data

```
# Frequency is null here because I already grouped data above
# I've also removed the argument for the log.path which will default
# to a temporary directory created by R which is deleted upon exiting an R session
result <- multimodal.fitting(dataPSD.BMI,
```

```
    frequency = NULL,
    labeling = T,
    max.iterations = 30,
    max.modes = 6,
    smoothing = F,
    lower.limit = 10,
    upper.limit = 1500,
    NMRSE.threshold = 0.05,
    FVU.threshold = 5,
    FVU.tolerance = 0.1,
    verbose = T)
```

```
## [1] "Log Path: /var/folders/2j/rmf9p1l50wg904hz8qkpvc4m0000gn/T//RtmptrTruY/multimodal20231031181523"
## [1] "Current Dataset Time: 2023-10-31 22:15:23 UTC"
## [1] "Dataset sampling frequency is 2.4 min"
## [1] "2023-10-31 22:15:23: Current Loop Iteration: 1, Remaining Variance: 94.93%, Number of Modes: 1"
## [1] "2023-10-31 22:15:23: Current Loop Iteration: 2, Remaining Variance: 3.93%, Number of Modes: 2"
## [1] "2023-10-31 22:15:23: Current Loop Iteration: 3, Remaining Variance: 2.57%, Number of Modes: 3"
## [1] "2023-10-31 22:15:23: Current Loop Iteration: 4, Remaining Variance: 1.63%, Number of Modes: 4"
## [1] "2023-10-31 22:15:23: Current Loop Iteration: 5, Remaining Variance: 1.34%, Number of Modes: 5"
## [1] "2023-10-31 22:15:23: Current Loop Iteration: 6, Remaining Variance: 1.3%, Number of Modes: 6"
```

```
## [1] "Concentration RMSE: 2132.57 n/cc"
```

```
# As there is only 1 data file for this set, we will flatten the list  
result <- flatten(result)
```

### 3.1.1 Outputs

The first element of result is a pass flag i.e. T or F

```
result$pass
```

```
## [1] TRUE
```

The second is the plot which consists of three panels

```
## Warning: Removed 50000 rows containing missing values or values outside the scale range  
## (`geom_ribbon()`).
```

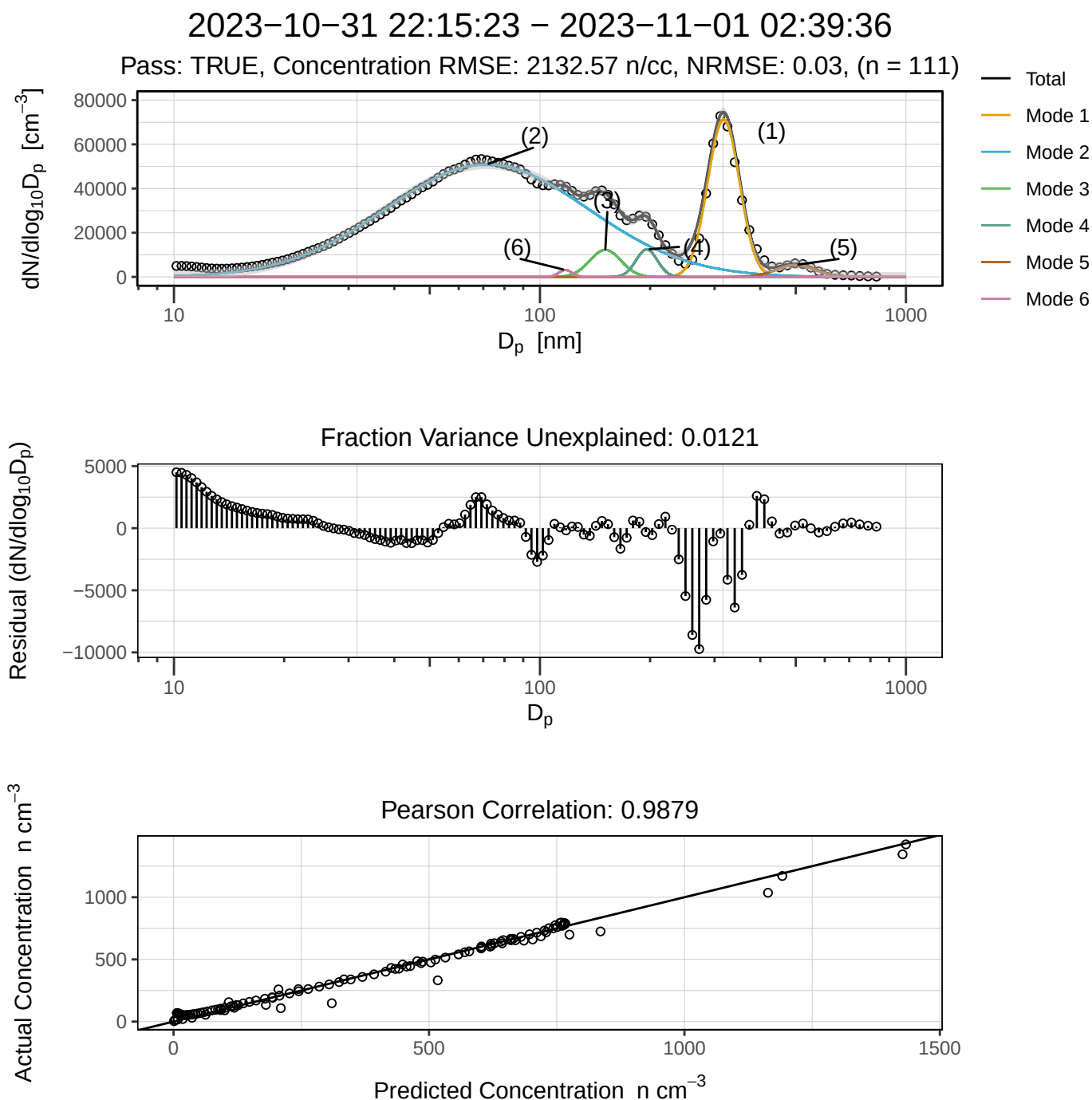


Figure S1 - An example of how to use the multimodal algorithm on example laboratory data. Note this reproduces the full version of Figure 2 in the manuscript.

The remaining outputs are predicted data, the data used to plot curves, and evaluation parameters

```
head(result[[3]])
```

| ##   | Dp    | Predicted dNdlogDp | Predicted dN | Actual dNdlogDp | Actual dN |
|------|-------|--------------------|--------------|-----------------|-----------|
| ## 1 | 10.16 | 453.91             | 6.301079     | 4952.512        | 68.74969  |
| ## 2 | 10.49 | 529.09             | 7.117198     | 4985.419        | 67.06273  |
| ## 3 | 10.82 | 612.31             | 8.465744     | 4903.316        | 67.79281  |
| ## 4 | 11.17 | 709.75             | 9.777611     | 4740.169        | 65.30121  |

```
## 5 11.53      820.21    11.251357      4507.124  61.82717
## 6 11.90      944.94    12.899759      4250.884  58.03054
##   Residual dNdlogDp Residual dN      Ratio
## 1      4498.602    62.44861 0.09165247
## 2      4456.329    59.94553 0.10612748
## 3      4291.006    59.32707 0.12487672
## 4      4030.419    55.52360 0.14973093
## 5      3686.914    50.57581 0.18198080
## 6      3305.944    45.13078 0.22229260
```

```
head(result[[4]])
```

```
##      Dp Mode 1 Mode 2 Mode 3 Mode 4 Mode 5 Mode 6 dNdlogDp
## 1 10.00      0 420.26      0      0      0      0 420.26
## 2 10.01      0 422.31      0      0      0      0 422.31
## 3 10.02      0 424.37      0      0      0      0 424.37
## 4 10.03      0 426.43      0      0      0      0 426.43
## 5 10.04      0 428.50      0      0      0      0 428.50
## 6 10.05      0 430.58      0      0      0      0 430.58
```

```
head(result[[5]])
```

```
##      Mode Label      N      GSD      Dpg      Max Lower Mode Upper Width
## 1      Mode 1 7709.7021 1.104493 317.61847 72801.737 249.85 311.03 451.56 13
## 2      Mode 2 35219.1406 1.888153 71.61388 53335.797 13.51 69.18 105.61 62
## 3      Mode 3 1299.9919 1.102360 150.90879 12544.712 126.94 147.69 172.59 8
## 4      Mode 4 906.6612 1.069269 195.75567 12079.366 165.93 194.62 272.39 12
## 5      Mode 5 715.3382 1.128914 499.31587 5631.458 430.34 497.83 830.76 13
## 6      Mode 6 140.9658 1.042290 117.38486 2934.445 98.26 117.86 136.85 9
##      BIC      RSS      TSS      R2 N T pval GSD T pval Dpg T pval
## 1 276.9295 150254441 8174258916 0.9816186 5e-12 2.3e-22 2.16e-22
## 2 1087.2011 88575112 19969539592 0.9955645 1e-65 1.1e-75 1.16e-73
## 3 152.9817 4783910 93721539 0.9489561 8e-07 2.7e-12 1.13e-12
## 4 262.9294 210120965 618259998 0.6601414 4e-03 3.1e-13 3.04e-13
## 5 212.3369 1489699 53034863 0.9719109 2e-10 2.0e-19 5.82e-20
## 6 179.0575 13928215 32875106 0.5763294 4e-02 2.1e-10 2.12e-10
```

```
result[[6]]
```

```
##      Pearson Correlation      RMSE      NRMSE dN RMSE      dN NRMSE Students T Test
## [1,]      0.9879 2132.57 0.02935411 37.15 0.02613263      0.8524
##      Chi-Squared
## [1,]      0.2391
```

## 3.2 Example 2 - Storm Peak Laboratory

```
## [1] "Log Path: ~/Library/CloudStorage/Box-Box/Multimodal Curve Fitting/log//multimodal20220323020341"
## [1] "Current Dataset Time: 2022-03-23 06:03:41 UTC"
## [1] "Dataset sampling frequency is 5 min"
## [1] "2022-03-23 06:03:41: Error, please modify lower and upper limits to accommodate data set"
```

Notice the failure message? This is because the dataset begins for bin diameter 9.14. Now we can retry with adjusted limits.

```
## [1] "Log Path: ~/Library/CloudStorage/Box-Box/Multimodal Curve Fitting/log//multimodal20220323020341"
## [1] "Current Dataset Time: 2022-03-23 06:03:41 UTC"
## [1] "Dataset sampling frequency is 5 min"
## [1] "2022-03-23 06:03:41: Current Loop Iteration: 1, Remaining Variance: 1.11%, Number of Modes: 1"
```

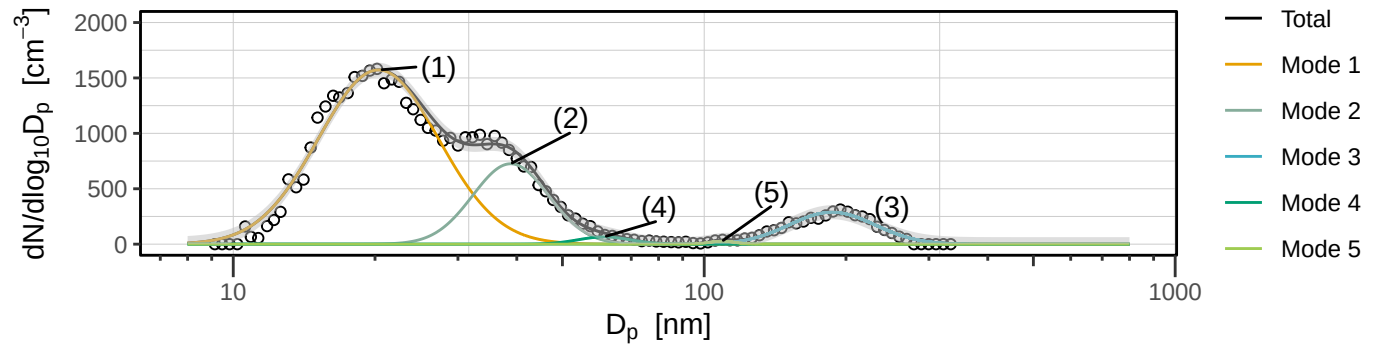
```
## [1] "2022-03-23 06:03:41: Current Loop Iteration: 2, Negative Concentration!"
## [1] "2022-03-23 06:03:41: Current Loop Iteration: 3, Negative Concentration!"
## [1] "2022-03-23 06:03:41: Current Loop Iteration: 4, Remaining Variance: 1.01%, Number of Modes: 2"
## [1] "2022-03-23 06:03:41: Current Loop Iteration: 5, Negative Concentration!"
## [1] "2022-03-23 06:03:41: Current Loop Iteration: 6, Negative Concentration!"
## [1] "2022-03-23 06:03:41: Terminating loop, no remaining peaks"
## [1] "Concentration RMSE: 391.48 n/cc"
```

For this file there is a NPF event, but currently the averaging across the entire day removes all temporal variation. We will instead select times between 07:00 and 15:00 and use an hourly frequency.

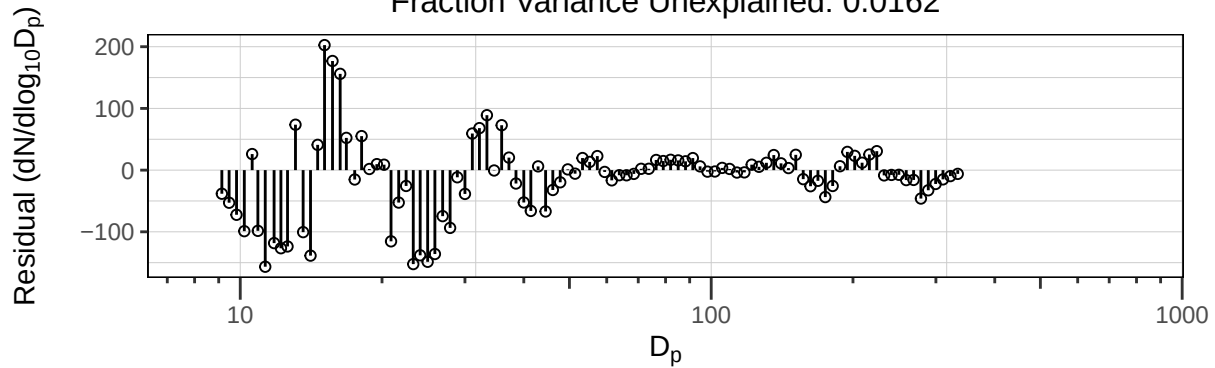
```
## [1] "Log Path: ~/Library/CloudStorage/Box-Box/Multimodal Curve Fitting/log//multimodal20220323070340."
## $`2022-03-23 11:00:00`
## Warning: Removed 70000 rows containing missing values or values outside the scale range
## (`geom_ribbon()`).
```

2022-03-23 11:03:40 – 2022-03-23 11:28:41

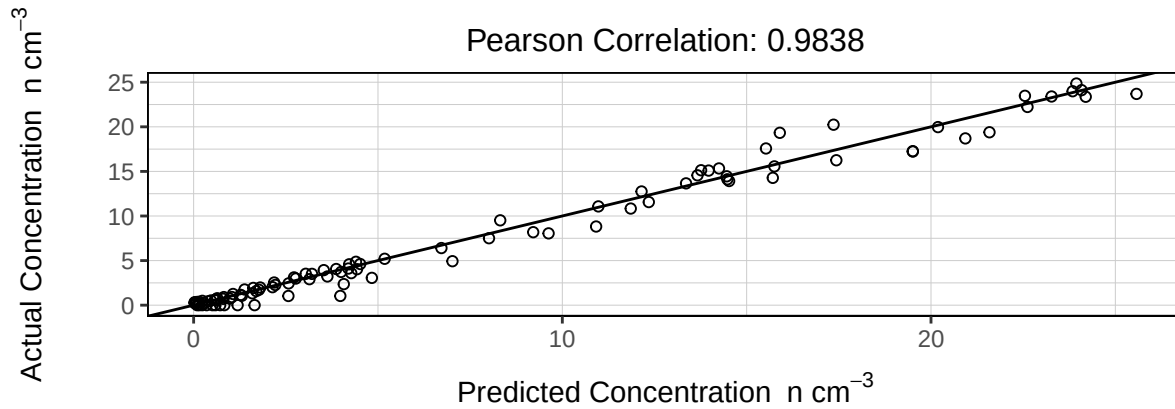
Pass: TRUE, Concentration RMSE: 63.42 n/cc, NRMSE: 0.04, (n = 6)



Fraction Variance Unexplained: 0.0162



Pearson Correlation: 0.9838



##

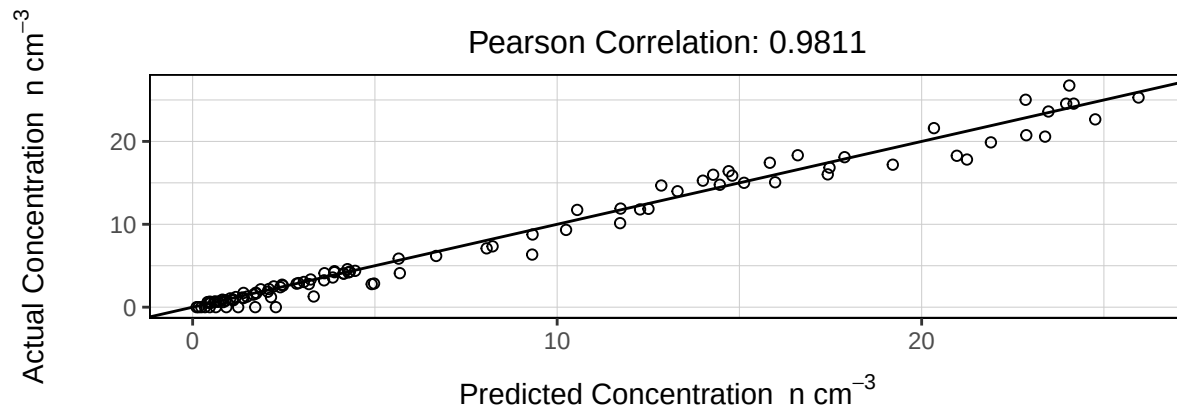
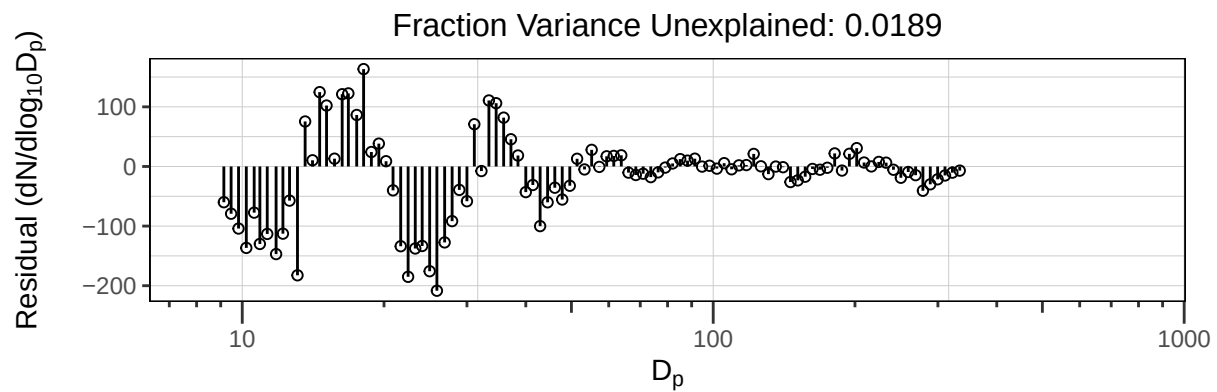
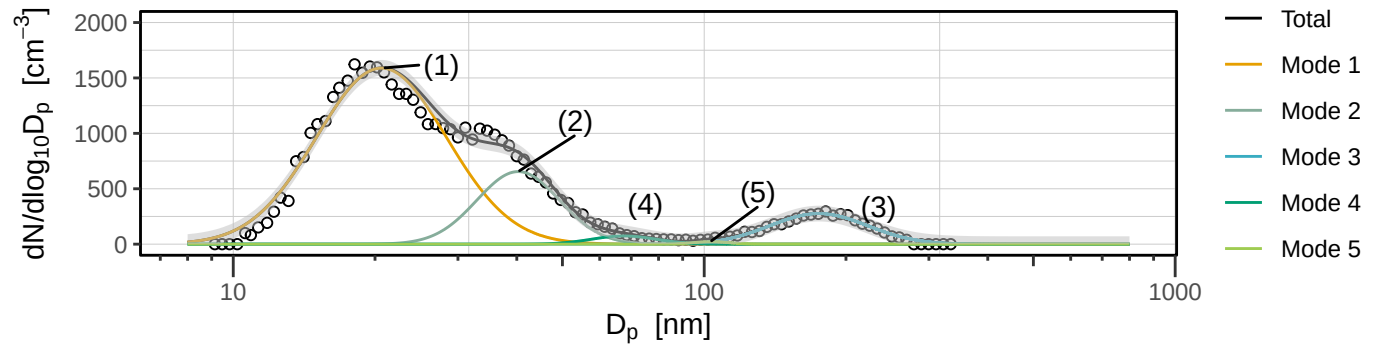
## \$`2022-03-23 12:00:00`

## Warning: Removed 70000 rows containing missing values or values outside the scale range  
## (`geom\_ribbon()`).



2022-03-23 11:33:41 – 2022-03-23 12:28:41

Pass: TRUE, Concentration RMSE: 71.02 n/cc, NRMSE: 0.04, (n = 12)



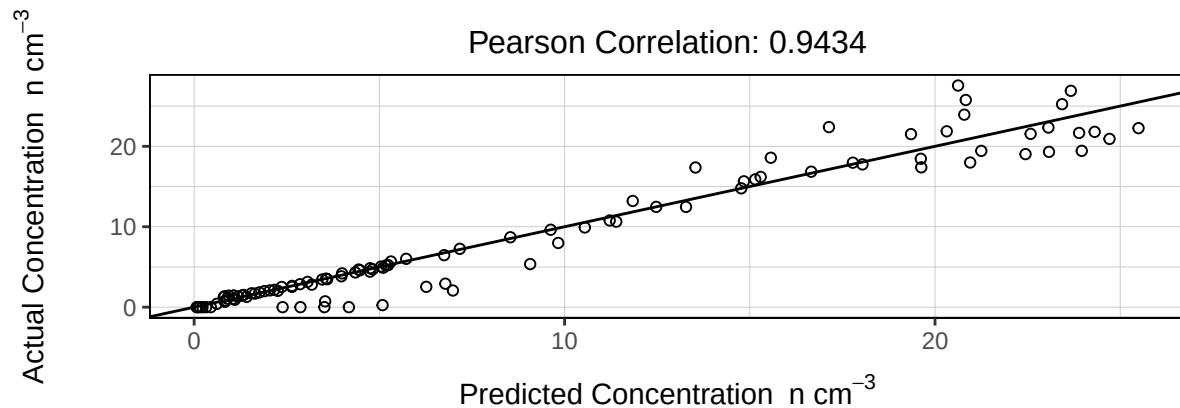
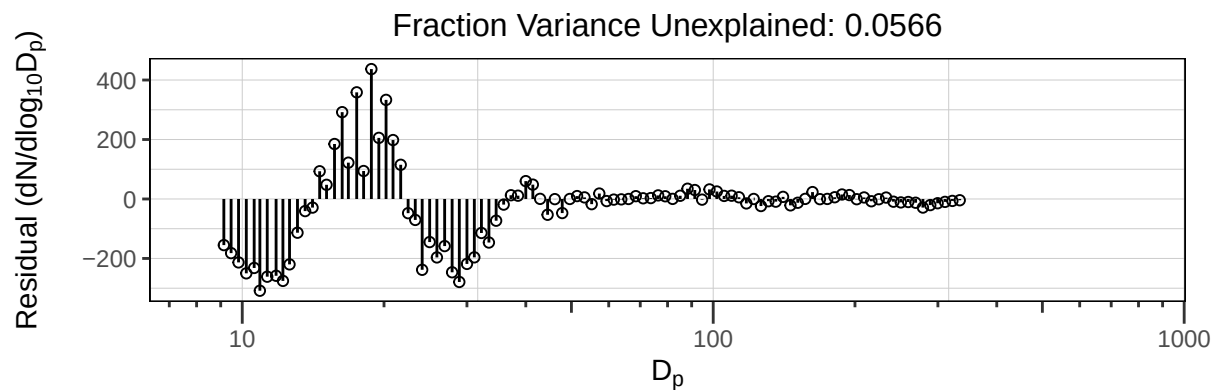
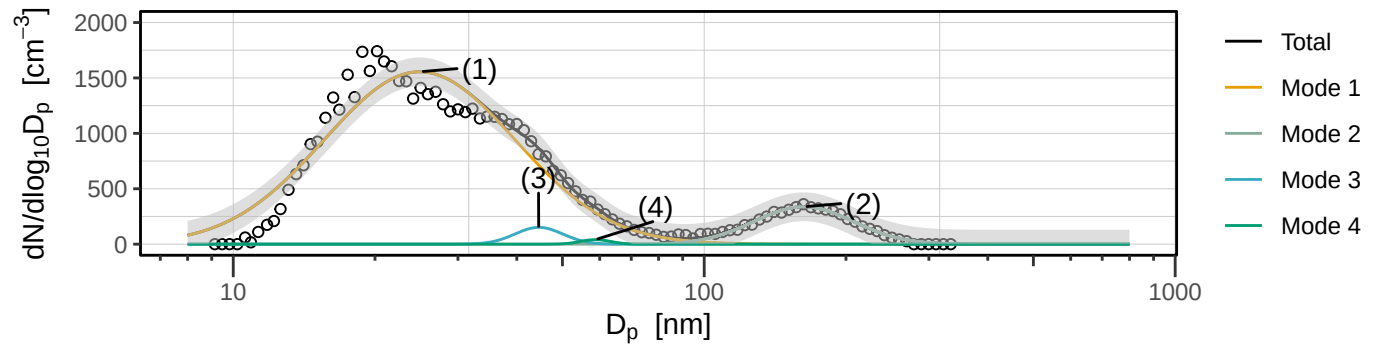
##

## \$`2022-03-23 13:00:00`

## Warning: Removed 70000 rows containing missing values or values outside the scale range  
## (`geom\_ribbon()`).

2022-03-23 12:33:41 – 2022-03-23 13:28:41

Pass: FALSE, Concentration RMSE: 129.85 n/cc, NRMSE: 0.07, (n = 12)



##

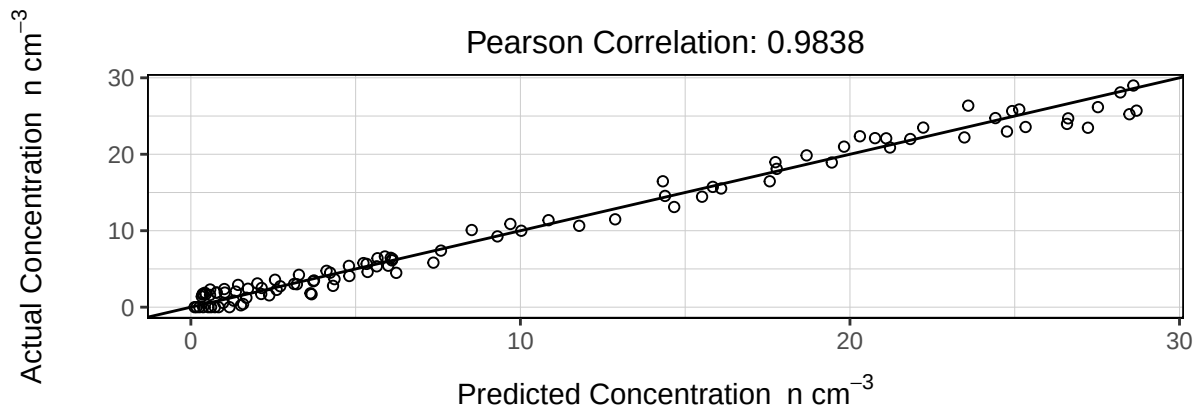
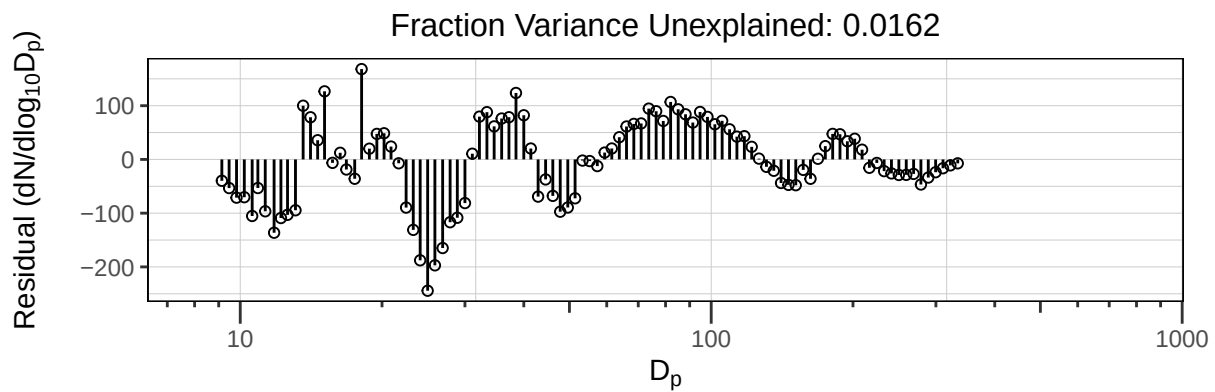
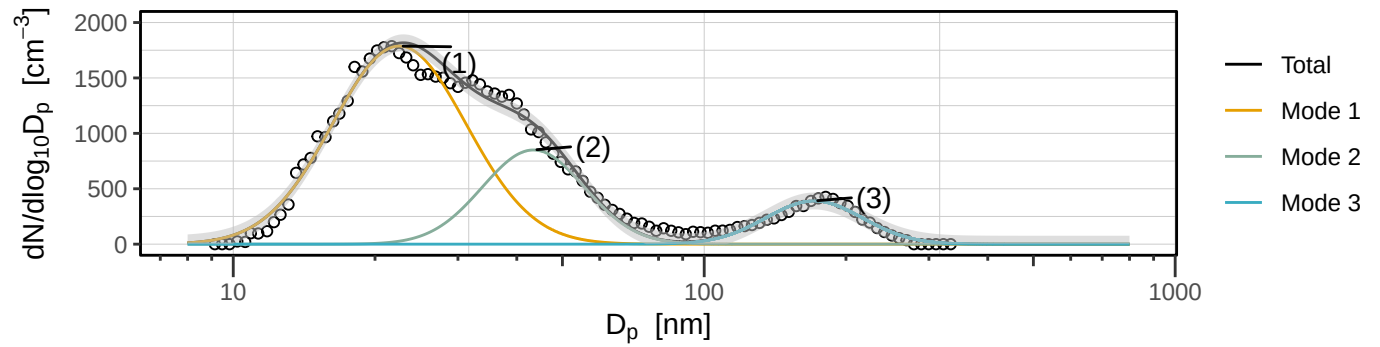
## \$`2022-03-23 14:00:00`

## Warning: Removed 70000 rows containing missing values or values outside the scale range

## (`geom\_ribbon()`).

2022-03-23 13:33:40 – 2022-03-23 14:28:41

Pass: TRUE, Concentration RMSE: 76.16 n/cc, NRMSE: 0.04, (n = 12)



##

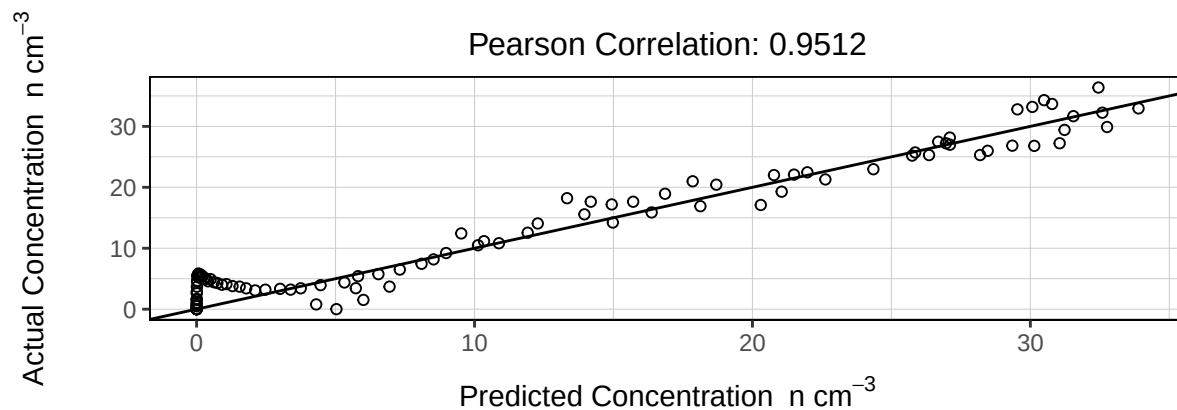
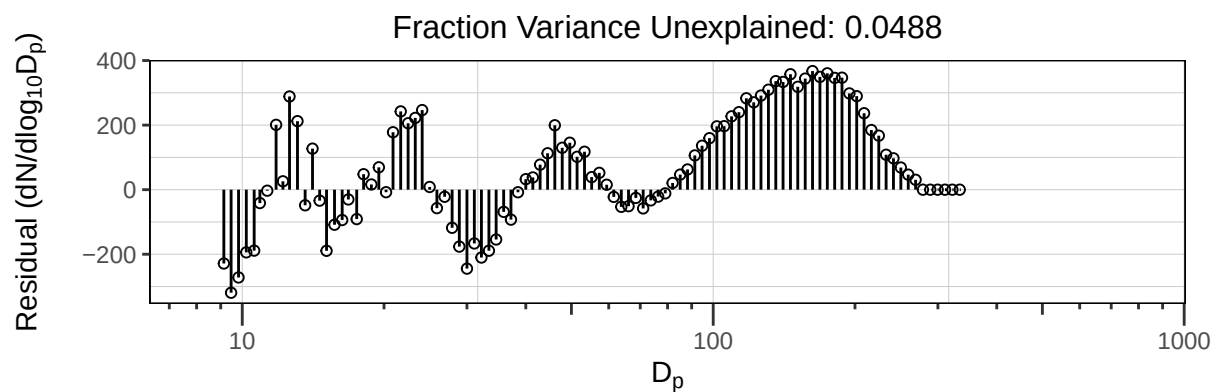
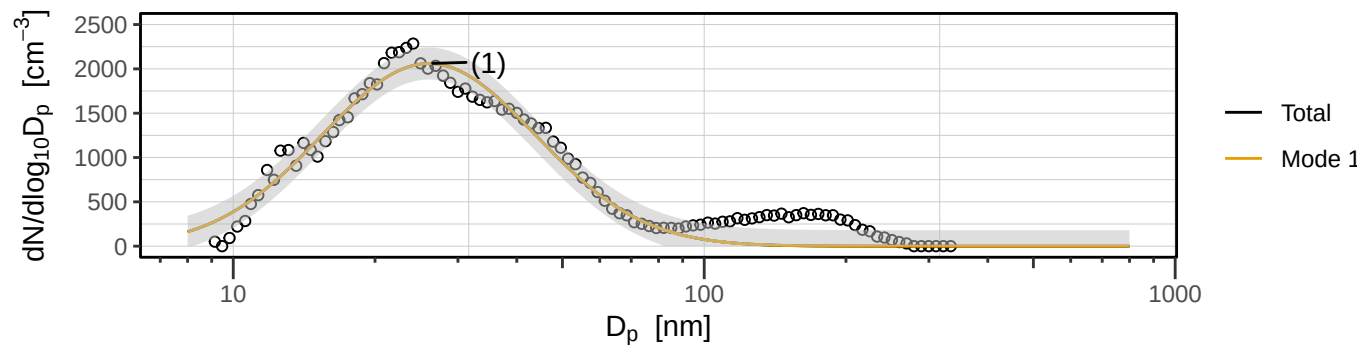
## \$`2022-03-23 15:00:00`

## Warning: Removed 70000 rows containing missing values or values outside the scale range

## (`geom\_ribbon()`).

2022-03-23 14:33:40 – 2022-03-23 15:28:40

Pass: FALSE, Concentration RMSE: 180.82 n/cc, NRMSE: 0.08, (n = 12)



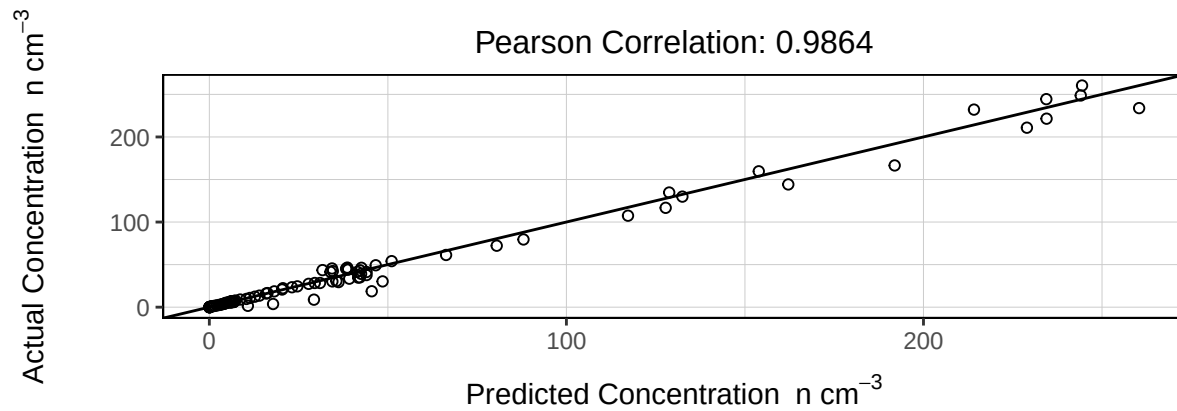
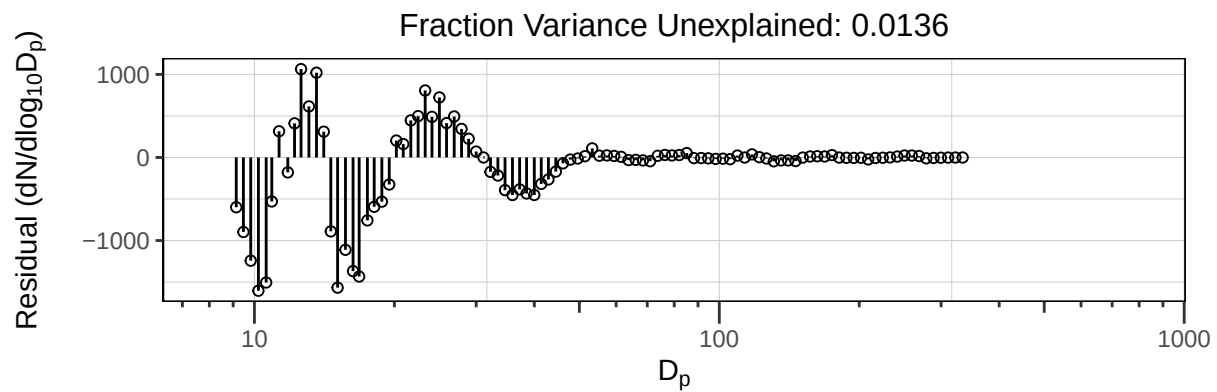
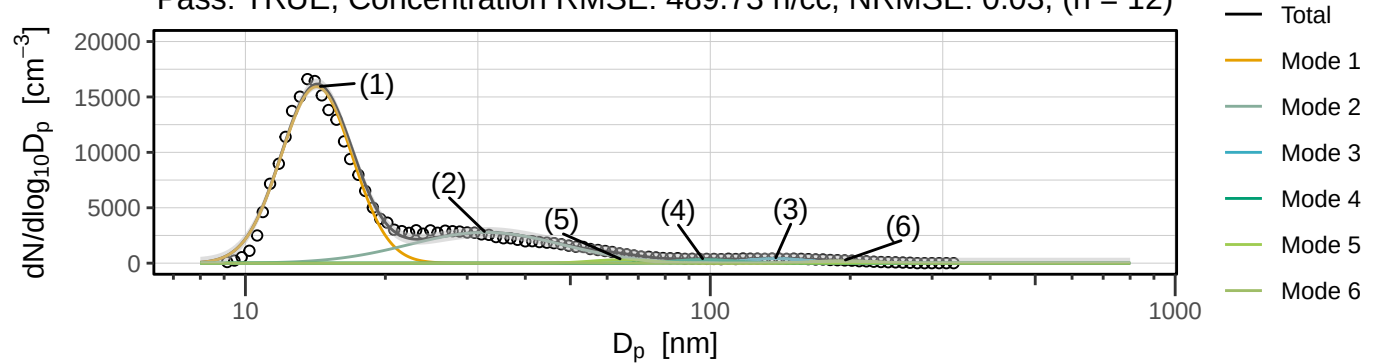
##

## \$`2022-03-23 16:00:00`

## Warning: Removed 70000 rows containing missing values or values outside the scale range  
## (`geom\_ribbon()`).

2022-03-23 15:33:40 – 2022-03-23 16:28:41

Pass: TRUE, Concentration RMSE: 489.73 n/cc, NRMSE: 0.03, (n = 12)



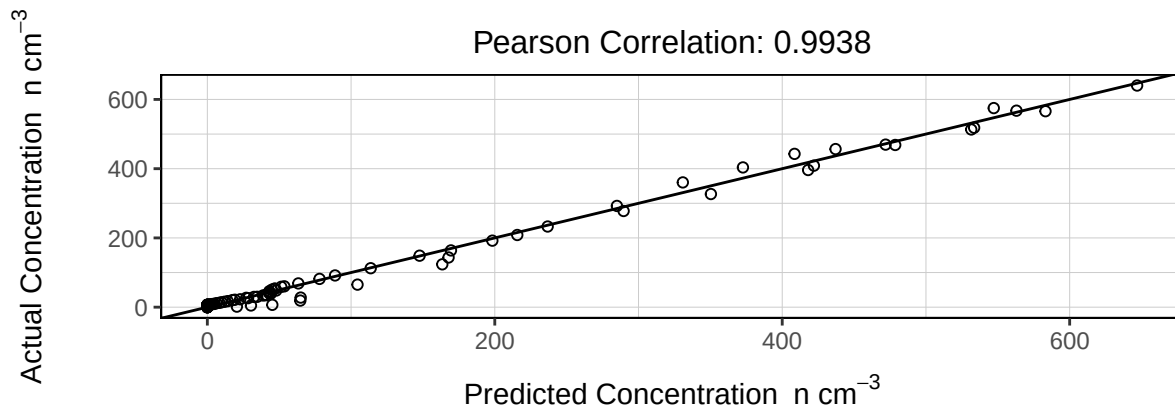
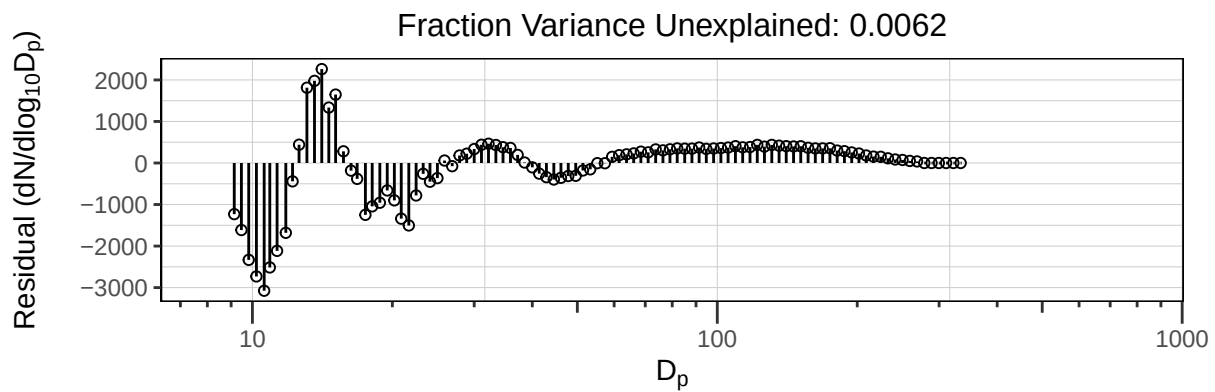
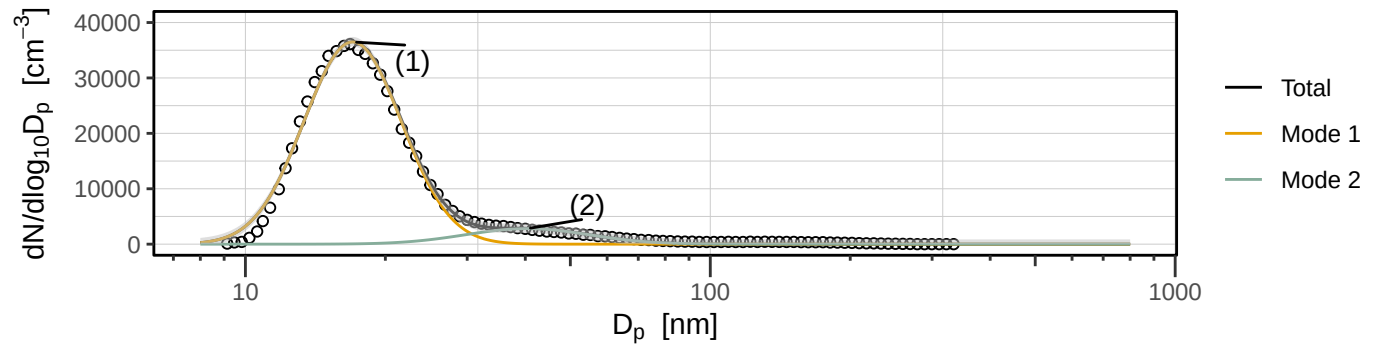
##

## \$`2022-03-23 17:00:00`

## Warning: Removed 70000 rows containing missing values or values outside the scale range  
## (`geom\_ribbon()`).

2022-03-23 16:33:41 – 2022-03-23 17:28:41

Pass: TRUE, Concentration RMSE: 852.68 n/cc, NRMSE: 0.02, (n = 12)



##

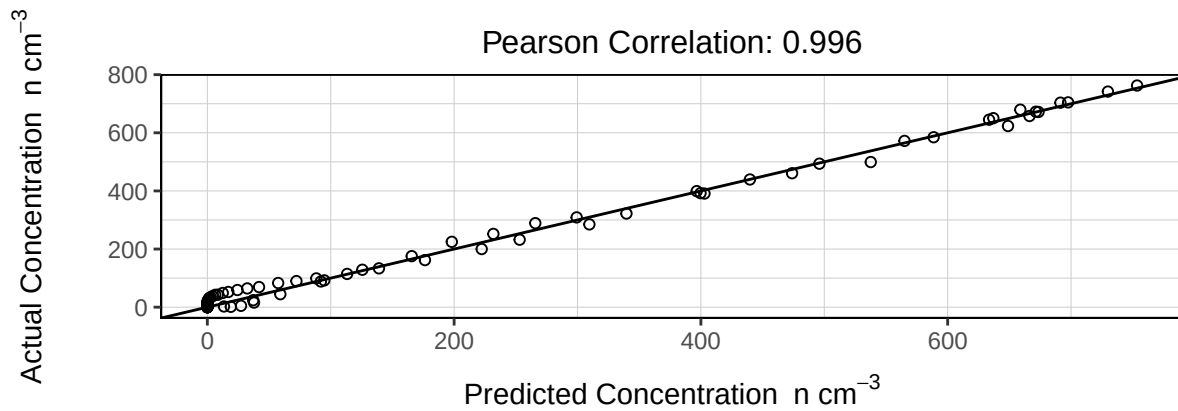
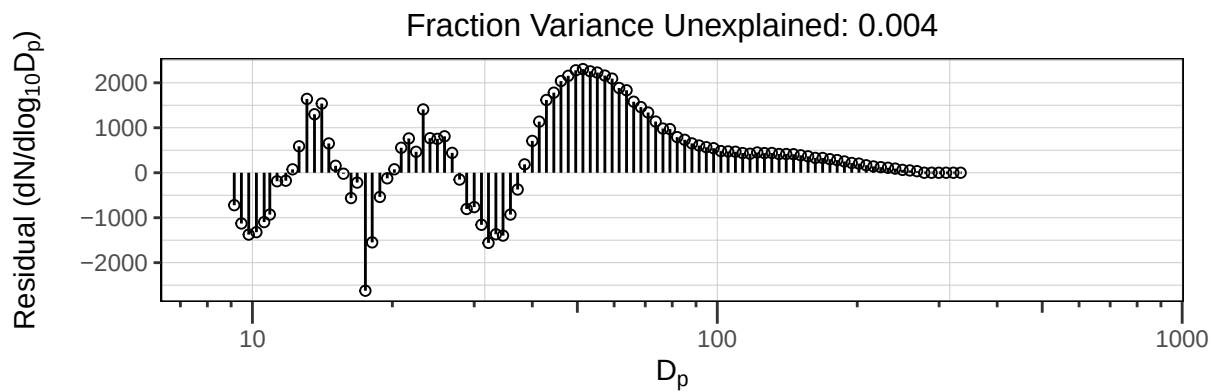
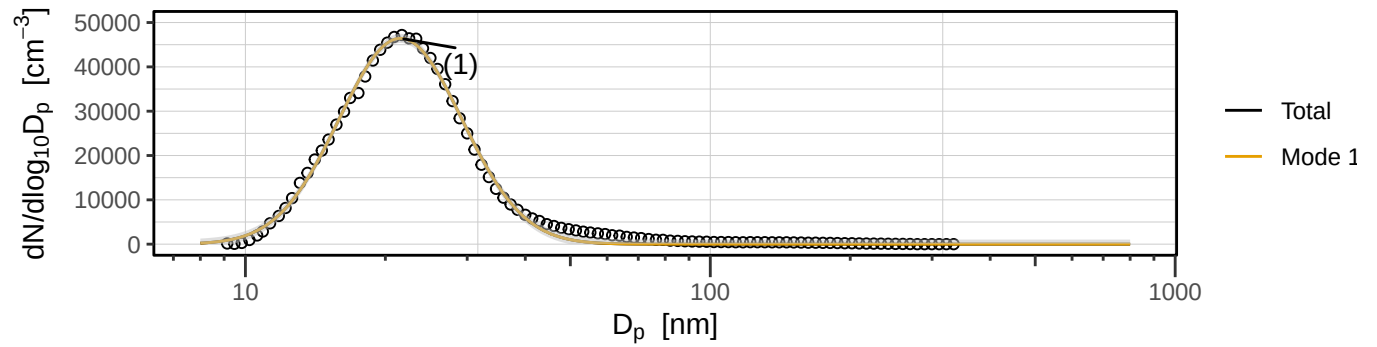
## \$`2022-03-23 18:00:00`

## Warning: Removed 70000 rows containing missing values or values outside the scale range

## (`geom\_ribbon()`).

2022-03-23 17:33:41 – 2022-03-23 18:28:40

Pass: TRUE, Concentration RMSE: 1035.03 n/cc, NRMSE: 0.02, (n = 12)



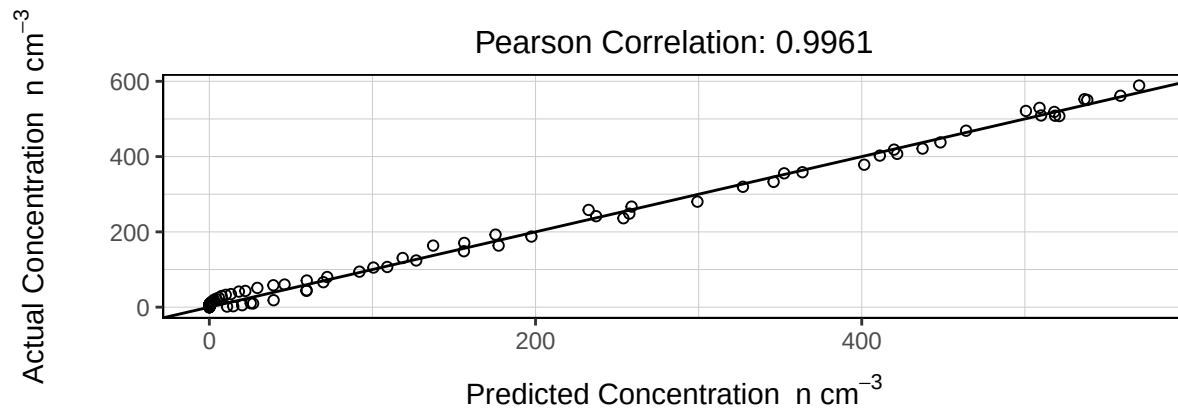
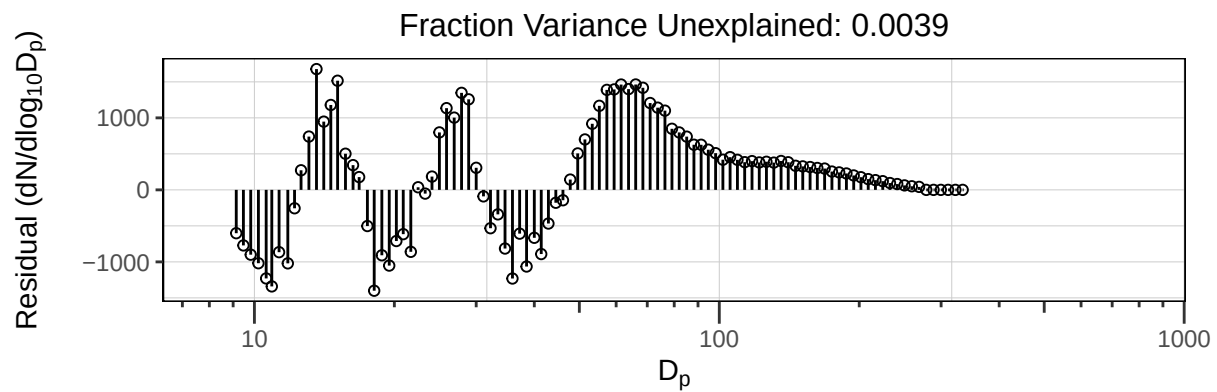
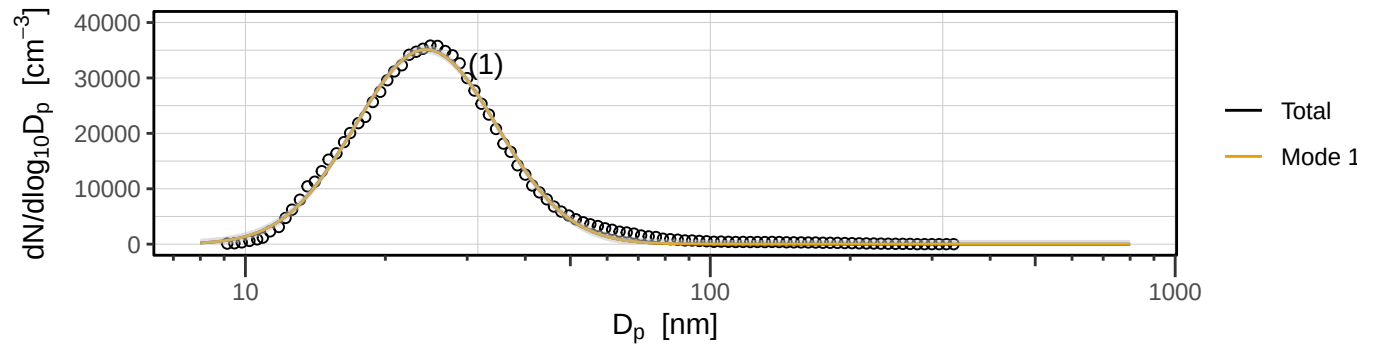
##

## \$`2022-03-23 19:00:00`

## Warning: Removed 70000 rows containing missing values or values outside the scale range  
## (`geom\_ribbon()`).

2022-03-23 18:33:40 – 2022-03-23 18:58:41

Pass: TRUE, Concentration RMSE: 764.08 n/cc, NRMSE: 0.02, (n = 6)



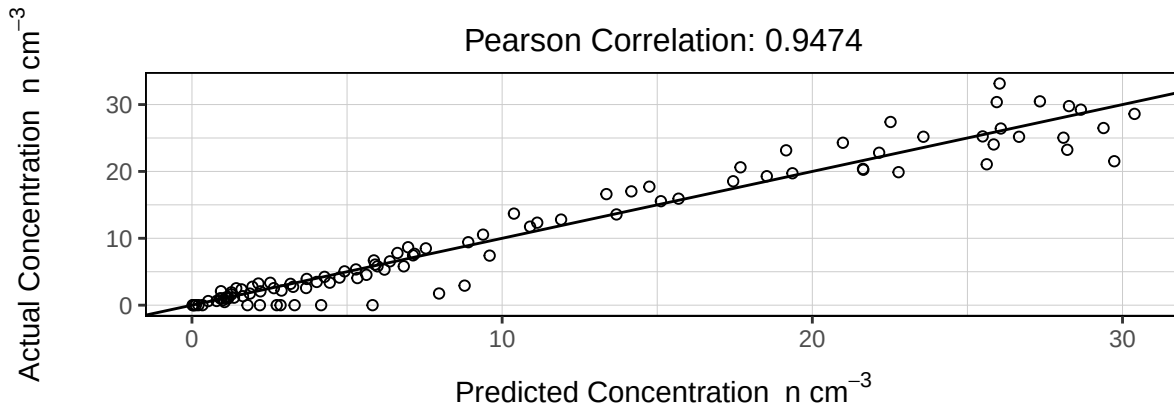
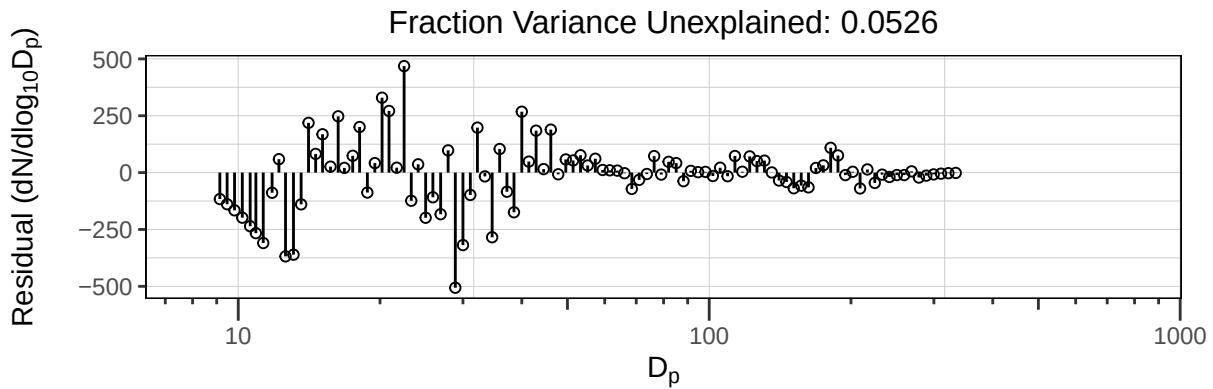
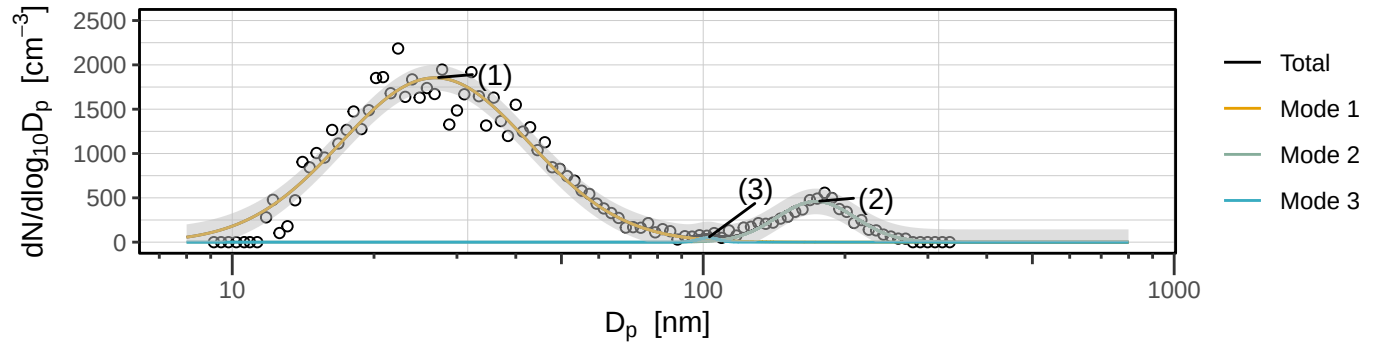
Note - for higher resolution scans, the variation within the scan may be too high to capture. The following is an example.

```
## [1] "Log Path: ~/Library/CloudStorage/Box-Box/Multimodal Curve Fitting/log//multimodal20220323100340."
## $`2022-03-23 14:05:00`
## Warning: Removed 70000 rows containing missing values or values outside the scale range
## (`geom_ribbon()`).
```



2022-03-23 14:03:40 – 2022-03-23 14:03:40

Pass: FALSE, Concentration RMSE: 144.36 n/cc, NRMSE: 0.07, (n = 1)



`head(dataPSD.BMI)`

| ##    | Local Time                                           | UTC Time            | #StartDate | StartTime    | EndDate |
|-------|------------------------------------------------------|---------------------|------------|--------------|---------|
| ##    | <POS>                                                | <POS>               | <int>      | <char>       | <int>   |
| ## 1: | 2023-10-31 18:15:23                                  | 2023-10-31 22:15:23 | 231031     | 18:15:23.757 | 231031  |
| ## 2: | 2023-10-31 18:17:47                                  | 2023-10-31 22:17:47 | 231031     | 18:17:47.862 | 231031  |
| ## 3: | 2023-10-31 18:20:11                                  | 2023-10-31 22:20:11 | 231031     | 18:20:11.965 | 231031  |
| ## 4: | 2023-10-31 18:22:36                                  | 2023-10-31 22:22:36 | 231031     | 18:22:36.078 | 231031  |
| ## 5: | 2023-10-31 18:25:00                                  | 2023-10-31 22:25:00 | 231031     | 18:25:00.193 | 231031  |
| ## 6: | 2023-10-31 18:27:24                                  | 2023-10-31 22:27:24 | 231031     | 18:27:24.307 | 231031  |
| ##    | EndTime StartSecs EndSecs Temp(K) Press(hPa) NumBins |                     |            | 10.16        | 10.49   |

```

##          <char>      <int>   <int>   <num>      <num>   <int>   <num>   <num>
## 1: 18:17:23.748      21      141  26.490    950.521    120 3420.58 3734.03
## 2: 18:19:47.858     165     285  26.466    950.415    120 3159.37 3015.60
## 3: 18:22:11.968     309     429  26.455    950.511    120 7291.97 8002.41
## 4: 18:24:36.083     453     573  26.435    950.417    120 7145.41 7368.59
## 5: 18:27:00.195     597     717  26.412    950.458    120 7046.85 7546.17
## 6: 18:29:24.306     742     862  26.409    950.604    120 2208.41 2063.78
##      10.82   11.17   11.53   11.90   12.28   12.68   13.09   13.51   13.94
##      <num>   <num>   <num>   <num>   <num>   <num>   <num>   <num>   <num>
## 1: 3931.78 4285.83 4785.94 5211.06 5268.06 4873.85 4272.60 3827.94 3760.99
## 2: 4087.27 4818.65 5183.06 5336.76 5346.21 5206.69 4998.48 4886.98 4944.24
## 3: 7251.37 6337.08 5376.83 4597.65 4178.56 4153.40 4396.36 4713.25 4961.93
## 4: 6523.60 5940.70 5641.67 5455.18 5259.88 5029.63 4781.73 4578.09 4466.42
## 5: 6556.28 5675.03 5022.23 4626.64 4408.62 4239.64 4075.71 3978.30 3973.91
## 6: 2907.14 3777.76 4616.84 5205.54 5305.98 4937.05 4398.17 4000.95 3931.87
##      14.40   14.86   15.34   15.84   16.35   16.88   17.43   17.99   18.57
##      <num>   <num>   <num>   <num>   <num>   <num>   <num>   <num>   <num>
## 1: 4106.44 4786.17 5649.61 6422.93 6777.77 6717.27 6708.59 7177.82 7989.92
## 2: 5049.88 5048.12 4945.73 4856.21 4877.03 5118.77 5683.88 6476.23 7140.36
## 3: 5073.22 5027.45 4903.05 4873.84 5070.76 5554.36 6386.57 7467.70 8367.67
## 4: 4424.22 4408.08 4513.98 4953.89 5805.99 6829.31 7542.95 7696.56 7641.66
## 5: 3991.92 3976.83 4005.68 4224.04 4727.91 5535.34 6520.87 7310.20 7577.44
## 6: 4208.82 4651.85 5008.29 5245.93 5548.47 6031.69 6648.09 7249.82 7612.15
##      19.18   19.80   20.45   21.11   21.80   22.51   23.25   24.01
##      <num>   <num>   <num>   <num>   <num>   <num>   <num>   <num>
## 1: 8548.47 8733.28 9211.15 10457.97 12012.68 12841.21 12669.16 12648.49
## 2: 7514.08 8066.04 9179.75 10343.19 11042.74 11890.38 13214.68 14357.91
## 3: 8732.85 8759.77 8792.58 9051.57 9983.58 12063.36 14492.67 15392.67
## 4: 7942.76 8795.73 9750.65 10256.04 10502.03 11103.43 12314.28 13682.46
## 5: 7667.47 8289.82 9573.20 11022.25 12169.90 13114.89 14118.46 14922.20
## 6: 7639.78 7705.24 8208.07 9024.83 9942.67 11218.32 12955.83 14377.35
##      24.79   25.61   26.45   27.31   28.21   29.14   30.10   31.10
##      <num>   <num>   <num>   <num>   <num>   <num>   <num>   <num>
## 1: 13986.12 16129.58 17227.65 17045.35 17272.03 18938.52 21684.38 23692.69
## 2: 15183.52 15898.18 16213.41 16797.10 18398.39 20293.99 21057.08 21117.08
## 3: 14657.90 14021.72 14551.29 15970.48 17511.27 19085.68 21431.74 23555.15
## 4: 15083.17 16876.54 18452.55 19452.92 20586.91 21894.61 22498.43 23233.30
## 5: 15221.22 15556.25 16922.90 19208.62 21415.54 23383.32 24749.90 24420.52
## 6: 14636.24 14337.90 14894.69 16789.12 19050.79 20696.30 21733.00 22751.64
##      32.13   33.19   34.29   35.43   36.61   37.83   39.10   40.41
##      <num>   <num>   <num>   <num>   <num>   <num>   <num>   <num>
## 1: 23181.99 22953.61 25254.42 27887.39 30067.61 32229.90 32671.72 32868.48
## 2: 23213.38 26595.63 27721.17 28408.51 29796.26 29939.27 31497.24 32851.97
## 3: 23757.11 24485.96 27037.65 29302.56 31065.86 32441.37 33273.56 34834.62
## 4: 25760.26 28023.02 27012.32 26622.12 30856.78 34567.88 34071.72 34225.12
## 5: 23573.56 24717.11 27277.00 28911.48 30296.23 32413.12 33819.28 34148.55
## 6: 24186.33 25943.83 28263.17 30151.38 30477.61 28989.20 27230.35 27257.49
##      41.76   43.17   44.62   46.13   47.69   49.31   50.98   52.72
##      <num>   <num>   <num>   <num>   <num>   <num>   <num>   <num>
## 1: 35450.00 38604.59 38953.14 39872.12 43913.64 45240.86 46239.88 47895.46
## 2: 32848.97 36334.22 38330.20 37851.80 39612.10 41868.67 43626.08 44671.76
## 3: 36850.88 38143.57 40412.24 41480.42 41738.99 42380.16 44951.39 47878.62
## 4: 35305.43 36483.99 40160.20 40913.30 41338.65 43762.25 45823.24 47474.99
## 5: 35653.22 39005.12 40350.96 41288.66 41667.12 41604.13 42420.06 44916.30

```

```

## 6: 29868.46 34058.13 38195.04 41527.81 40805.77 40652.88 41817.84 43309.53
##      54.52   56.39   58.33   60.34   62.43   64.59   66.84   69.18
##      <num>   <num>   <num>   <num>   <num>   <num>   <num>   <num>
## 1: 48281.58 50674.20 52528.16 50908.68 50761.38 49602.70 50528.73 52279.97
## 2: 45577.47 46407.03 46156.82 47093.80 49093.31 50892.83 50109.76 54196.07
## 3: 47918.56 47875.31 49415.92 48505.99 47098.50 48811.16 50357.77 51081.66
## 4: 48298.86 50519.05 53224.56 50885.04 47354.77 50035.90 51154.73 50535.06
## 5: 46813.32 49133.65 49786.21 51540.26 50666.08 54057.69 57616.11 54979.51
## 6: 44099.67 44213.12 43783.88 49945.58 53104.36 52352.23 55076.31 51120.83
##      71.60   74.12   76.74   79.46   82.29   85.24   88.30   91.49
##      <num>   <num>   <num>   <num>   <num>   <num>   <num>   <num>
## 1: 50352.99 48915.17 49036.42 47895.21 45138.51 42063.32 41833.62 41722.89
## 2: 53978.99 50493.50 48187.22 44805.96 43748.55 43132.88 41691.31 39140.19
## 3: 52832.49 50063.21 47126.57 47528.52 45807.05 46552.90 45463.66 43382.42
## 4: 49539.02 49384.89 47012.91 47039.31 48432.25 47355.53 43283.66 42326.72
## 5: 54723.19 54407.00 54407.02 51001.54 47072.40 48224.70 42988.93 42028.12
## 6: 53844.64 54517.47 47695.84 48707.64 49135.27 45931.29 44152.33 41760.10
##      94.80   98.26  101.86  105.61  109.52  113.60  117.86  122.30
##      <num>   <num>   <num>   <num>   <num>   <num>   <num>   <num>
## 1: 39969.75 35038.72 32446.79 32587.17 30219.92 33671.06 33991.86 31547.75
## 2: 37680.04 31840.54 34578.53 32434.63 36885.01 36445.42 34801.38 31460.89
## 3: 38850.88 36612.29 36634.99 35406.63 34316.29 35955.79 32048.60 30834.88
## 4: 41253.58 37798.44 36200.26 34462.05 33798.13 35368.46 35001.89 31128.95
## 5: 39703.71 36910.56 34052.27 36365.87 38649.88 35738.99 35325.53 31006.84
## 6: 36851.33 35329.81 35153.62 33348.54 37888.60 35243.59 34359.17 34295.79
##      126.94  131.79  136.85  142.15  147.69  153.49  159.56  165.93
##      <num>   <num>   <num>   <num>   <num>   <num>   <num>   <num>
## 1: 27432.75 28020.87 31142.28 32206.87 33388.63 31906.87 27556.64 21712.79
## 2: 27405.51 26126.54 27194.69 31262.51 31840.73 29377.42 25640.83 20579.79
## 3: 30856.82 27673.98 28863.90 30873.96 33176.87 33095.38 27237.82 20676.43
## 4: 31438.97 28104.94 28422.75 31550.96 32987.19 33022.69 26099.18 21490.21
## 5: 31225.77 28995.79 30330.49 32069.83 32074.74 33180.79 28002.80 21229.33
## 6: 34255.16 31391.69 32262.48 35523.92 36315.22 34053.78 30792.82 24445.10
##      172.59  179.59  186.92  194.62  202.71  211.21  220.14  229.54
##      <num>   <num>   <num>   <num>   <num>   <num>   <num>   <num>
## 1: 19280.14 20543.98 20651.60 19030.09 16269.33 12139.91  8732.44 4561.59
## 2: 20773.91 21115.16 20380.08 19801.21 18840.94 14031.73  9938.76 6253.27
## 3: 21683.90 23658.16 25327.43 23842.45 19942.16 16333.91 12696.32 7650.79
## 4: 19329.21 20022.26 20520.26 22393.30 20091.52 15331.79 10676.07 6659.61
## 5: 21184.50 22315.74 24783.85 24756.46 19801.98 15676.01 10671.15 6076.28
## 6: 20811.07 23654.21 23256.48 25116.91 18546.70 13154.91 11376.66 8653.37
##      239.44  249.85  260.82  272.39  284.59  297.45  311.03  325.37
##      <num>   <num>   <num>   <num>   <num>   <num>   <num>   <num>
## 1: 2873.65 3003.38 5532.21 17202.01 40744.48 60474.39 73052.66 67931.15
## 2: 3858.99 3109.18 5901.98 18149.35 39104.25 62219.36 72902.43 68268.13
## 3: 3905.74 3666.30 7276.22 20093.63 37902.27 61023.07 74393.81 66100.27
## 4: 4263.19 4177.95 7186.87 16972.94 40544.63 61328.13 70400.53 65854.64
## 5: 4027.26 4348.13 6070.95 17347.98 37584.21 61411.90 73498.74 67506.95
## 6: 5734.11 4079.36 6781.74 17969.24 40607.33 63940.04 77308.55 68125.98
##      340.52  356.53  373.46  391.36  410.30  430.34  451.56  474.03  497.83
##      <num>   <num>   <num>   <num>   <num>   <num>   <num>   <num>   <num>
## 1: 48048.68 31763.35 18487.14  9178.38 5363.63 3381.23 3259.09 4331.26 5111.66
## 2: 47841.06 30071.61 16786.55  9125.37 5836.74 3570.41 2698.44 3306.43 4627.42
## 3: 46664.94 32120.27 17603.53 10592.59 5481.44 3070.19 2597.24 3497.24 3810.38

```

```
## 4: 47402.11 33812.84 19663.36 9710.91 5916.86 3928.15 2984.95 3203.95 4999.13
## 5: 49807.95 31643.16 18237.92 9933.42 6239.27 2752.77 3720.83 4089.52 4663.98
## 6: 48837.00 33299.19 19291.92 11030.60 6025.11 3592.56 4514.76 5593.17 4698.33
##      523.04 549.76 578.09 608.12 639.96 673.74 709.57 747.59 787.94 830.76
##      <num> <num> <num> <num> <num> <num> <num> <num> <num> <num>
## 1: 3996.63 3495.25 2421.18 1041.61 542.10 457.46 490.19 136.88 34.25 157.63
## 2: 4165.94 3149.61 1438.42 1009.10 601.28 536.62 359.14 235.85 224.78 62.12
## 3: 3624.72 3009.17 1071.64 587.13 234.54 398.96 410.90 213.38 332.68 76.72
## 4: 3769.93 3194.02 1893.51 914.24 547.04 365.28 169.72 221.69 199.39 40.96
## 5: 3670.90 2798.97 1762.13 449.38 586.24 773.44 398.76 164.58 80.64 134.43
## 6: 4794.93 2871.07 1847.37 956.91 858.19 510.44 478.79 310.43 381.14 36.52
##      876.23 NumFlag      Units Size Range
##      <num> <int> <char> <num>
## 1: 77.70      0 dNdlogDp      866.07
## 2: 58.30      0 dNdlogDp      866.07
## 3: 57.90      0 dNdlogDp      866.07
## 4:  5.39      0 dNdlogDp      866.07
## 5: 18.44      0 dNdlogDp      866.07
## 6:  3.90      0 dNdlogDp      866.07
```

```
head(dataPSD.TSI)
```

```
##           Local Time           UTC Time Sample #      Date Start Time
##           <POS<           <POS<      <int>      <char>      <char>
## 1: 2022-03-23 00:03:41 2022-03-23 06:03:41      22042 03/23/22      00:03:41
## 2: 2022-03-23 00:08:40 2022-03-23 06:08:40      22043 03/23/22      00:08:40
## 3: 2022-03-23 00:13:40 2022-03-23 06:13:40      22044 03/23/22      00:13:40
## 4: 2022-03-23 00:18:41 2022-03-23 06:18:41      22045 03/23/22      00:18:41
## 5: 2022-03-23 00:23:41 2022-03-23 06:23:41      22046 03/23/22      00:23:41
## 6: 2022-03-23 00:28:41 2022-03-23 06:28:41      22047 03/23/22      00:28:41
##      Sample Temp(C) Sample Pressure(kPa) Mean Free Path(m) Gas Viscosity(Pa*s)
##      <num>           <num>           <num>           <num>
## 1:      21.8           65.0           1.042800e-07      1.826716e-05
## 2:      21.8           64.9           1.044406e-07      1.826716e-05
## 3:      21.7           65.0           1.042511e-07      1.826237e-05
## 4:      21.8           65.0           1.042961e-07      1.826716e-05
## 5:      21.6           65.0           1.042061e-07      1.825759e-05
## 6:      21.8           65.1           1.042319e-07      1.826716e-05
##      Diameter Midpoint  9.14  9.47  9.82 10.2      10.6      10.9      11.3      11.8
##      <lgcl> <num> <num> <num> <num> <num> <num> <num> <num>
## 1:      NA      0      0      0      0      0.000 197.349 165.190      0.000
## 2:      NA      0      0      0      0      0.000      0.000      0.000 139.885
## 3:      NA      0      0      0      0      0.000      0.000      0.000      0.000
## 4:      NA      0      0      0      0      0.000      0.000      0.000 139.686
## 5:      NA      0      0      0      0 239.221      0.000 165.063 139.563
## 6:      NA      0      0      0      0      0.000 197.246      0.000      0.000
##      12.2      12.6      13.1      13.6      14.1      14.6      15.1      15.7
##      <num> <num> <num> <num> <num> <num> <num> <num>
## 1: 119.149 205.893 537.7770 1100.4600 731.200 1264.100 1272.090 1045.530
## 2:  0.000  0.000  89.7529  865.7820 695.574  534.916  817.785 1075.470
## 3: 238.234 205.841 896.0800   78.5859 972.367  827.438  696.483  846.219
## 4: 476.667 308.884 448.2090 393.0710 555.828  864.461  536.244  910.998
## 5: 238.134 308.638 716.5930 549.9040 555.448  802.190  870.332 1057.760
## 6: 238.190 102.902 627.1480 892.1420 600.387 1278.010  845.519  946.997
##      16.3      16.8      17.5      18.1      18.8      19.5      20.2      20.9
```

```

##      <num>      <num>      <num>      <num>      <num>      <num>      <num>      <num>
## 1:  945.511    705.087    750.964    1029.850    1136.800    1121.590    798.448    811.939
## 2:  585.939    662.720    853.130    1087.340    1063.430    978.252    824.356    778.284
## 3:  810.294    746.247    825.571    1014.800    788.091    714.964    755.608    822.320
## 4: 1497.180    1022.380    1423.590    947.111    1008.230    818.397    800.194    931.906
## 5: 1266.740    852.461    1021.970    1116.670    1166.100    1222.520    971.384    1035.850
## 6: 1169.030    746.401    924.198    1007.570    900.127    993.537    1021.950    538.486
##      21.7      22.5      23.3      24.1      25.0      25.9      26.9      27.9      28.9
##      <num>      <num>      <num>      <num>      <num>      <num>      <num>      <num>      <num>
## 1: 552.977    936.257    772.250    753.779    970.894    991.501    821.234    912.458    994.252
## 2: 686.882    787.487    885.758    771.547    1064.100    516.136    673.900    963.984    852.055
## 3: 893.441    948.449    942.425    746.502    739.741    982.018    874.202    1071.140    1079.850
## 4: 748.017    534.817    780.669    896.600    986.982    870.489    989.264    893.928    1139.280
## 5: 847.582    796.664    942.216    864.329    818.518    901.165    894.329    1050.530    1009.270
## 6: 793.556    892.247    710.593    817.135    759.351    1218.100    1062.580    1030.360    881.614
##      30.0      31.1      32.2      33.4      34.6      35.9      37.2      38.5
##      <num>      <num>      <num>      <num>      <num>      <num>      <num>      <num>
## 1: 1101.620    1059.470    1051.010    1110.000    874.964    772.121    923.254    830.352
## 2:  810.815    982.204    1129.770    1156.650    1147.190    1030.990    927.985    740.158
## 3:  814.663    1103.450    1130.750    995.607    1097.650    1117.610    871.307    764.166
## 4: 1222.570    824.732    1097.810    896.415    898.849    909.217    1095.970    714.966
## 5: 1164.170    1033.240    893.356    911.370    1023.610    1028.900    778.422    642.334
## 6:  904.722    940.746    883.367    1218.180    1000.030    989.585    751.879    784.413
##      40.0      41.4      42.9      44.5      46.1      47.8      49.6      51.4      53.3
##      <num>      <num>      <num>      <num>      <num>      <num>      <num>      <num>      <num>
## 1: 704.077    543.343    532.085    413.761    340.885    192.353    203.852    149.168    199.1210
## 2: 589.793    597.713    440.377    322.472    258.539    266.625    214.363    199.551    160.5590
## 3: 655.200    517.612    335.643    361.668    248.236    249.078    149.799    114.920    124.5500
## 4: 803.899    539.732    422.166    350.520    360.424    266.994    97.3948    193.785    105.5690
## 5: 663.602    580.939    464.785    251.628    277.625    220.313    169.569    212.348    109.7310
## 6: 647.183    477.660    506.529    327.103    286.743    264.709    214.734    159.438    79.7374
##      55.2      57.3      59.4      61.5      63.8      66.1      68.5      71.0      73.7
##      <num>      <num>      <num>      <num>      <num>      <num>      <num>      <num>      <num>
## 1:  97.9716    106.3630    154.9910    153.8300    145.934    233.5010    232.727    184.466    155.427
## 2: 146.3700    112.1270    137.2040    118.8250    122.510    80.3680    152.510    103.625    209.588
## 3: 105.6750    54.4646    101.7230    104.1780    152.435    134.3620    196.023    186.525    139.293
## 4:  93.0872    121.1920    81.2171    99.5188    141.156    94.9131    223.259    135.609    184.474
## 5:  74.0632    131.2950    132.4830    115.4640    175.797    71.8226    167.940    169.304    141.137
## 6: 126.9810    134.5350    113.8760    126.8410    109.639    123.8490    128.060    108.854    278.049
##      76.4      79.1      82.0      85.1      88.2      91.4      94.7      98.2      101.8
##      <num>      <num>      <num>      <num>      <num>      <num>      <num>      <num>      <num>
## 1: 160.150    190.965    210.829    258.099    297.700    253.898    288.489    330.759    301.084
## 2: 152.934    234.367    185.909    185.675    245.477    339.798    348.205    333.769    382.287
## 3: 151.405    241.164    232.721    235.929    248.013    329.950    258.245    407.918    293.789
## 4: 134.784    215.345    229.982    320.966    335.539    328.506    239.511    310.813    301.674
## 5: 187.564    254.811    249.418    340.641    278.084    221.111    205.900    296.925    257.330
## 6: 148.768    231.010    165.701    228.159    256.833    301.819    242.626    262.261    279.860
##      105.5     109.4     113.4     117.6     121.9     126.3     131.0     135.8     140.7
##      <num>      <num>      <num>      <num>      <num>      <num>      <num>      <num>      <num>
## 1: 365.322    304.885    315.507    288.473    305.216    231.640    195.869    192.862    249.229
## 2: 350.268    326.903    391.164    310.955    336.524    325.825    337.661    283.029    295.786
## 3: 325.951    370.892    273.395    309.865    265.613    261.247    320.133    288.952    284.883
## 4: 283.761    310.761    297.287    348.271    267.395    312.350    319.894    226.978    297.472
## 5: 319.653    334.636    303.913    309.348    275.987    301.593    247.608    227.335    250.395

```

```

## 6: 393.031 245.679 301.759 252.979 342.495 265.423 250.865 321.399 233.386
##      145.9   151.2   156.8   162.5   168.5   174.7   181.1   187.7   194.6
##      <num>  <num>  <num>  <num>  <num>  <num>  <num>  <num>  <num>
## 1: 260.072 201.171 204.748 175.090 238.277 136.991 111.7630 138.5130 83.9202
## 2: 226.807 230.971 237.514 202.594 146.391 153.754 165.2710 92.9477 107.3920
## 3: 229.934 188.236 195.393 181.404 105.471 167.408 127.0000 156.6190 98.8299
## 4: 262.732 249.206 235.344 233.248 185.762 167.274 148.6380 95.8195 115.1420
## 5: 185.565 232.205 198.869 164.970 137.844 158.655 92.3299 92.2701 98.3238
## 6: 255.239 207.533 171.502 119.280 205.706 153.636 126.0880 117.4170 107.6480
##      201.7   209.1   216.7   224.7   232.9   241.4   250.3   259.5   269.0
##      <num>  <num>  <num>  <num>  <num>  <num>  <num>  <num>  <num>
## 1: 109.3220 121.7440 33.8008 81.0674 35.7504 20.1837 31.8373 15.79130 5.33835
## 2: 104.8160 71.4689 71.1985 33.6003 20.4320 25.8819 10.4397 15.79370 16.01750
## 3: 98.0943 81.5530 58.8758 52.8620 23.3219 35.7559 16.1332 5.26366 16.01470
## 4: 80.5776 108.6790 77.5540 28.2422 21.6584 39.6153 21.4253 8.87974 17.68660
## 5: 92.7493 73.5486 82.7527 38.1945 15.3207 20.7012 25.0776 22.07870 5.33805
## 6: 68.0021 43.5719 65.3531 47.8294 18.1787 31.0521 20.8752 10.52700 4.46073
##      278.8 289.0 299.6 310.6 322.0 333.8 346.0 Scan Up Time(s) Retrace Time(s)
##      <int> <int> <int> <int> <int> <int> <int>          <int>          <int>
## 1:      0      0      0      0      0      0      0          280          10
## 2:      0      0      0      0      0      0      0          280          10
## 3:      0      0      0      0      0      0      0          280          10
## 4:      0      0      0      0      0      0      0          280          10
## 5:      0      0      0      0      0      0      0          280          10
## 6:      0      0      0      0      0      0      0          280          10
##      Down Scan First Scans Per Sample Impactor Type(cm) Sheath Flow(lpm)
##      <lgcl>          <int>          <num>          <int>
## 1:      FALSE              1          0.0457          10
## 2:      FALSE              1          0.0457          10
## 3:      FALSE              1          0.0457          10
## 4:      FALSE              1          0.0457          10
## 5:      FALSE              1          0.0457          10
## 6:      FALSE              1          0.0457          10
##      Aerosol Flow(lpm) CPC Inlet Flow(lpm) CPC Sample Flow(lpm) Low Voltage
##      <int>          <int>          <int>          <num>
## 1:              1              1              1      10.6730
## 2:              1              1              1      10.6568
## 3:              1              1              1      10.6732
## 4:              1              1              1      10.6714
## 5:              1              1              1      10.6749
## 6:              1              1              1      10.6779
##      High Voltage Lower Size(nm) Upper Size(nm) Density(g/cc) Title Status Flag
##      <num>          <num>          <num>          <num> <lgcl>      <char>
## 1:      9855.99      8.97687      352.269          1.2      NA Normal Scan
## 2:      9848.78      8.97687      352.269          1.2      NA Normal Scan
## 3:      9854.71      8.97687      352.269          1.2      NA Normal Scan
## 4:      9855.27      8.97687      352.269          1.2      NA Normal Scan
## 5:      9854.15      8.97687      352.269          1.2      NA Normal Scan
## 6:      9858.15      8.97687      352.269          1.2      NA Normal Scan
##      td(s)   tf(s) D50(nm) Median(nm) Mean(nm) Geo.Mean(nm) Mode(nm)
##      <num>  <num>  <num>    <num>    <num>    <num>    <num>
## 1: 1.779 2.25624 265.182    28.1484 43.6574    32.2018 15.1247
## 2: 1.779 2.25624 265.018    30.2327 46.9183    34.8112 33.3762
## 3: 1.779 2.25624 265.163    29.0698 45.3157    33.6499 32.1968

```

```
## 4: 1.779 2.25624 265.166    28.6246 45.0059    33.1420 16.2531
## 5: 1.779 2.25624 265.160    27.3360 42.4647    31.5314 16.2531
## 6: 1.779 2.25624 265.231    27.7849 43.3304    32.1639 14.5902
##      Geo.Std. Dev. Total Conc.(#/cc) Comment      Units Size Range
##      <num>          <num> <lgcl>    <char>      <num>
## 1:      2.04694          638.399    NA dNdlogDp    336.86
## 2:      2.04897          589.268    NA dNdlogDp    336.86
## 3:      2.03663          598.152    NA dNdlogDp    336.86
## 4:      2.05389          626.208    NA dNdlogDp    336.86
## 5:      2.02267          634.772    NA dNdlogDp    336.86
## 6:      2.02913          615.324    NA dNdlogDp    336.86
```

```
head(dataPSD.NAS)
```

```
##      starttime      endtime pressure (hPa) temperature (K)  9.14
## 1 2022-03-22 00:00:00 2022-03-22 01:00:00      646.33      299.72 474.24
## 2 2022-03-22 01:00:00 2022-03-22 01:59:59      646.83      299.25 84.12
## 3 2022-03-22 01:59:59 2022-03-22 03:00:00      646.58      298.76 0.00
## 4 2022-03-22 03:00:00 2022-03-22 04:00:00      647.00      298.46 0.00
## 5 2022-03-22 04:00:00 2022-03-22 04:59:59      646.83      298.20 0.00
## 6 2022-03-22 04:59:59 2022-03-22 06:00:00      647.08      297.82 0.00
##      9.47    9.82    10.2    10.6    10.9    11.3    11.8    12.2    12.6
## 1 957.28 2867.13 3583.47 5114.35 6310.64 7191.34 7549.04 8580.99 8822.77
## 2 298.20 916.45 1245.15 1485.14 2468.77 2897.71 3331.94 3617.07 4289.60
## 3 120.74 189.00 529.25 605.65 796.45 1144.47 1584.89 1746.31 2045.49
## 4 0.00 94.34 133.73 75.24 198.43 312.59 320.85 275.00 400.82
## 5 0.00 31.36 25.14 40.92 168.13 238.74 299.34 346.10 351.32
## 6 0.00 0.00 25.07 122.19 151.01 210.47 278.15 342.93 542.03
##      13.1    13.6    14.1    14.6    15.1    15.7    16.3    16.8    17.5
## 1 7952.76 8010.82 8162.86 7990.85 7451.82 7584.54 7593.25 7594.88 7001.90
## 2 4424.07 4447.83 4597.11 4500.57 4620.33 4802.66 4865.29 4819.59 5048.73
## 3 2123.54 2227.72 2054.91 2302.66 2327.27 2574.78 2489.49 2572.97 2945.02
## 4 441.17 583.74 626.55 646.45 752.25 845.95 818.85 881.72 1049.40
## 5 475.65 438.01 612.68 584.59 726.27 692.28 805.44 880.78 923.34
## 6 468.29 766.36 774.15 798.08 919.40 972.99 1031.76 1092.83 1195.97
##      18.1    18.8    19.5    20.2    20.9    21.7    22.5    23.3    24.1
## 1 7125.60 7862.48 8124.80 8159.64 8364.12 8668.43 9002.40 9461.98 9622.86
## 2 5513.54 5543.39 5945.24 6631.56 6552.99 7130.38 7386.48 7614.73 7806.07
## 3 3080.90 3204.40 3573.71 3759.66 4030.89 4260.40 4499.01 4618.08 4902.01
## 4 1049.61 1210.46 1189.02 1250.51 1195.23 1263.97 1262.21 1328.49 1260.05
## 5 997.21 1153.40 1169.85 1226.61 1239.19 1358.63 1385.03 1447.41 1496.10
## 6 1343.78 1458.61 1407.93 1530.73 1582.33 1734.30 1774.63 1756.64 1859.80
##      25      25.9    26.9    27.9    28.9    30      31.1    32.2
## 1 9946.65 10164.08 10489.97 10751.10 10815.08 10605.00 10928.09 10570.33
## 2 8382.28 8610.70 8977.47 8990.91 9017.20 9184.89 9058.58 8908.22
## 3 5232.13 5463.19 5566.40 5534.47 5708.35 5517.94 5529.11 5402.95
## 4 1305.21 1373.60 1372.72 1341.94 1378.23 1410.03 1323.68 1346.79
## 5 1459.50 1536.34 1602.51 1622.83 1670.06 1636.87 1661.84 1692.39
## 6 1908.20 2064.13 2066.88 2153.87 2184.21 2167.84 2311.34 2311.05
##      33.4    34.6    35.9    37.2    38.5    40      41.4    42.9    44.5
## 1 10358.50 9783.52 9459.79 8558.29 7695.16 6851.74 5834.16 4942.83 4192.25
## 2 8438.41 8037.41 7532.57 7088.88 6175.92 5608.94 4724.73 4034.35 3385.69
## 3 4980.05 4867.38 4367.58 3831.24 3402.95 2974.85 2467.29 2100.91 1787.02
## 4 1428.47 1320.27 1200.93 1190.69 1150.40 1097.74 982.27 910.10 800.29
## 5 1676.72 1646.94 1566.61 1531.97 1458.84 1378.62 1266.97 1189.41 1029.79
```

```

## 6 2285.59 2436.51 2321.06 2303.95 2236.54 2069.25 1980.24 1747.04 1582.02
##      46.1    47.8    49.6    51.4    53.3    55.2    57.3    59.4    61.5    63.8
## 1 3382.34 2716.13 2182.88 1793.04 1464.58 1209.39 998.06 848.05 734.84 592.79
## 2 2784.91 2355.69 1772.08 1507.75 1253.15 1031.68 854.03 758.11 595.86 523.25
## 3 1423.36 1228.62 971.92 828.83 676.32 537.27 482.39 431.73 326.24 292.78
## 4 754.27 612.65 572.04 513.85 407.18 363.87 281.49 258.54 228.20 192.09
## 5 900.15 769.38 664.89 586.86 527.89 403.64 344.26 297.05 259.85 248.29
## 6 1388.00 1162.01 1037.57 901.58 715.39 633.97 520.10 452.40 379.16 336.34
##      66.1    68.5     71    73.7    76.4    79.1     82    85.1    88.2    91.4    94.7
## 1 512.72 445.46 378.32 318.03 290.38 284.55 252.50 260.34 251.02 273.03 214.79
## 2 449.11 415.56 338.35 324.20 283.20 251.22 217.34 224.30 231.22 236.16 241.40
## 3 247.59 220.45 185.24 181.72 189.66 174.19 179.05 163.21 191.91 181.62 226.92
## 4 171.30 169.45 163.22 155.08 177.27 178.84 179.88 182.70 171.43 184.97 215.76
## 5 217.41 197.53 161.24 159.57 178.57 168.03 160.96 162.04 184.80 183.87 181.94
## 6 280.22 261.75 232.72 202.27 217.25 203.28 195.96 203.14 209.34 204.90 221.08
##      98.2   101.8   105.5   109.4   113.4   117.6   121.9   126.3    131   135.8   140.7
## 1 258.38 259.50 256.63 268.92 259.48 258.36 280.98 289.10 302.29 298.64 321.96
## 2 244.99 241.06 253.38 269.55 286.54 308.75 283.43 329.53 308.09 333.24 350.34
## 3 192.55 212.90 217.24 215.47 232.49 232.15 248.19 245.20 264.20 252.14 262.13
## 4 204.90 207.46 223.79 208.16 201.93 246.67 232.52 236.46 265.16 293.46 278.25
## 5 176.06 216.26 226.13 227.97 230.01 247.26 247.21 256.99 296.23 264.36 287.18
## 6 244.08 234.17 265.16 264.20 260.05 271.70 287.46 316.44 309.72 320.14 308.20
##      145.9   151.2   156.8   162.5   168.5   174.7   181.1   187.7   194.6   201.7   209.1
## 1 335.77 353.06 328.67 362.97 360.43 363.98 350.93 346.04 332.69 320.46 298.03
## 2 329.74 352.83 367.66 350.14 357.34 364.43 327.49 334.80 338.99 293.30 288.43
## 3 290.72 276.36 318.40 294.14 301.60 310.36 297.91 285.13 275.97 266.37 233.51
## 4 281.34 291.47 291.63 275.51 286.84 307.70 264.72 247.35 218.87 204.69 159.03
## 5 273.42 293.82 296.25 292.59 307.69 297.63 286.49 273.43 247.65 197.44 170.20
## 6 311.86 322.37 339.79 318.80 334.50 342.54 300.79 297.08 250.80 249.10 211.09
##      216.7   224.7   232.9   241.4   250.3   259.5   269   278.8   289   299.6   310.6
## 1 261.44 243.50 190.16 151.17 123.31 92.10 72.33      0    0      0    0
## 2 249.12 208.33 170.19 147.77 115.52 91.64 74.70      0    0      0    0
## 3 210.80 169.81 144.35 118.50 88.52 80.92 59.73      0    0      0    0
## 4 143.56 110.82 90.40 78.33 53.16 38.25 30.68      0    0      0    0
## 5 150.52 120.76 101.64 68.88 58.16 44.93 36.39      0    0      0    0
## 6 168.26 141.73 117.16 85.99 76.30 55.52 42.99      0    0      0    0
## numflag, no unit
## 1      0
## 2      0
## 3      0
## 4      0
## 5      0
## 6      0

```

```
head(dataPSD.NC)
```

```

##           Time    10.55    10.9    11.3    11.75    12.2    12.65    13.1
## 1 2023-07-19 00:00:01 229.408 548.848 262.211 884.073 530.819 794.500 763.913
## 2 2023-07-19 00:05:00 114.746 299.476 218.582 461.397 597.981 531.017 921.694
## 3 2023-07-19 00:10:00 230.551 451.271 526.986 502.107 579.764 938.770 849.747
## 4 2023-07-19 00:15:00 115.809 352.599 397.037 310.385 445.339 1003.796 908.580
## 5 2023-07-19 00:20:00 114.704 249.476 305.913 768.758 543.062 753.482 818.479
## 6 2023-07-19 00:25:00 347.944 353.094 309.220 388.476 411.583 639.520 854.460
##      13.55    14.05    14.6    15.15    15.7    16.25    16.85    17.5
## 1 1121.157 1130.514 1211.083 1389.714 1453.444 1709.798 1621.780 1876.027

```



|      |          |          |          |          |          |          |          |          |
|------|----------|----------|----------|----------|----------|----------|----------|----------|
| ## 2 | 736.839  | 1214.537 | 1172.016 | 1166.224 | 1471.679 | 1399.218 | 1331.886 | 1919.116 |
| ## 3 | 964.090  | 993.853  | 1014.759 | 1132.285 | 1231.431 | 1539.507 | 1618.466 | 1654.262 |
| ## 4 | 1106.781 | 1338.346 | 1124.499 | 1145.662 | 1598.000 | 1301.520 | 1320.362 | 1795.484 |
| ## 5 | 682.445  | 1086.181 | 929.296  | 1274.957 | 1257.640 | 1441.239 | 1507.432 | 1662.602 |
| ## 6 | 738.635  | 1046.237 | 923.729  | 1389.188 | 1166.476 | 1297.628 | 1539.155 | 1552.144 |
| ##   | 18.1     | 18.75    | 19.45    | 20.15    | 20.9     | 21.7     | 22.5     | 23.3     |
| ## 1 | 1764.448 | 2429.017 | 2329.730 | 2320.435 | 2426.032 | 2144.505 | 2953.922 | 2892.823 |
| ## 2 | 2078.902 | 1784.736 | 2238.934 | 2599.956 | 2238.237 | 2729.816 | 2568.009 | 2865.300 |
| ## 3 | 1978.002 | 1924.727 | 2113.580 | 2389.900 | 2136.968 | 2457.808 | 2574.584 | 2803.770 |
| ## 4 | 1702.826 | 2441.895 | 1930.326 | 2193.416 | 2250.921 | 2491.292 | 2514.423 | 2778.490 |
| ## 5 | 1730.383 | 2237.354 | 1960.943 | 2203.576 | 2214.458 | 2177.672 | 2251.208 | 2567.292 |
| ## 6 | 1842.513 | 2022.649 | 1957.764 | 2218.728 | 2282.920 | 2335.917 | 2350.628 | 2576.894 |
| ##   | 24.15    | 25.05    | 25.95    | 26.9     | 27.9     | 28.9     | 29.95    | 31.05    |
| ## 1 | 3023.508 | 3295.483 | 3538.897 | 3728.871 | 4002.189 | 4419.072 | 4430.606 | 4916.476 |
| ## 2 | 2921.435 | 3019.030 | 3655.349 | 3642.867 | 4121.817 | 4351.931 | 4344.164 | 4808.162 |
| ## 3 | 2863.671 | 2955.779 | 3165.180 | 3597.900 | 3835.769 | 4206.578 | 4491.183 | 4772.361 |
| ## 4 | 2950.127 | 3041.181 | 3033.596 | 3525.306 | 3764.251 | 4518.231 | 4226.971 | 4906.754 |
| ## 5 | 2842.133 | 2832.622 | 3096.053 | 3368.349 | 3519.172 | 4116.027 | 4131.004 | 4908.662 |
| ## 6 | 3193.599 | 2849.140 | 3278.841 | 3570.402 | 3492.563 | 3708.533 | 4139.758 | 4553.236 |
| ##   | 32.2     | 33.4     | 34.6     | 35.85    | 37.2     | 38.55    | 39.95    | 41.45    |
| ## 1 | 5183.347 | 5586.184 | 5627.987 | 5668.352 | 6305.616 | 5850.432 | 5525.757 | 4909.097 |
| ## 2 | 4940.063 | 5348.845 | 5788.506 | 6273.896 | 6252.516 | 5840.920 | 5754.282 | 5006.534 |
| ## 3 | 5333.142 | 5527.131 | 5553.507 | 6158.542 | 6319.847 | 6255.833 | 5876.435 | 5583.143 |
| ## 4 | 5182.285 | 5376.875 | 5981.352 | 6303.903 | 6453.669 | 6093.425 | 6367.504 | 6294.676 |
| ## 5 | 5283.231 | 5743.592 | 5555.021 | 6240.056 | 6650.838 | 6758.908 | 6682.508 | 6402.453 |
| ## 6 | 5036.475 | 5258.476 | 6059.448 | 6698.119 | 6786.079 | 7332.025 | 7485.635 | 7316.705 |
| ##   | 42.95    | 44.5     | 46.15    | 47.85    | 49.6     | 51.4     | 53.25    | 55.2     |
| ## 1 | 4626.498 | 3861.933 | 3801.762 | 3259.180 | 3058.614 | 2875.609 | 3132.449 | 3321.056 |
| ## 2 | 4518.195 | 3963.110 | 3345.568 | 3221.533 | 2945.356 | 3083.876 | 3290.623 | 3431.242 |
| ## 3 | 4825.924 | 4435.265 | 4056.036 | 3627.474 | 3482.259 | 3395.029 | 3427.702 | 3578.296 |
| ## 4 | 5848.510 | 5260.221 | 4673.045 | 4295.160 | 3913.575 | 3708.291 | 3756.823 | 3770.353 |
| ## 5 | 6157.749 | 5815.503 | 4749.304 | 4416.361 | 4045.551 | 3734.515 | 3659.251 | 3759.374 |
| ## 6 | 6902.182 | 6340.164 | 5786.006 | 5161.751 | 4537.692 | 3820.945 | 3991.784 | 3805.736 |
| ##   | 57.25    | 59.35    | 61.5     | 63.75    | 66.1     | 68.55    | 71.05    | 73.65    |
| ## 1 | 3756.458 | 4388.521 | 4840.560 | 5370.918 | 5533.685 | 5802.646 | 5955.583 | 5348.883 |
| ## 2 | 3676.774 | 4480.724 | 4916.737 | 5357.544 | 5836.761 | 6005.729 | 5812.702 | 5457.548 |
| ## 3 | 3809.919 | 4494.913 | 4679.721 | 5719.344 | 5860.879 | 6114.155 | 5822.916 | 5876.069 |
| ## 4 | 4157.233 | 4528.021 | 4867.136 | 5398.713 | 5804.219 | 6060.933 | 5897.384 | 6031.204 |
| ## 5 | 3965.716 | 4435.867 | 4905.022 | 5539.349 | 5624.933 | 5966.754 | 6137.991 | 5788.471 |
| ## 6 | 4081.439 | 4503.414 | 4585.815 | 5276.229 | 5883.960 | 6018.710 | 5708.181 | 5926.518 |
| ##   | 76.35    | 79.15    | 82.05    | 85.05    | 88.2     | 91.45    | 94.8     | 98.25    |
| ## 1 | 5257.314 | 4937.997 | 4372.268 | 4001.376 | 3502.182 | 2896.232 | 2984.634 | 2654.561 |
| ## 2 | 5160.905 | 4876.296 | 4356.225 | 3935.288 | 3555.791 | 3108.744 | 2785.382 | 2623.546 |
| ## 3 | 5419.656 | 4842.824 | 4316.584 | 4053.013 | 3371.008 | 3159.801 | 2864.063 | 2725.578 |
| ## 4 | 5485.083 | 5166.305 | 4567.694 | 4043.328 | 3539.274 | 3090.468 | 2876.918 | 2712.535 |
| ## 5 | 5460.610 | 5072.785 | 4481.935 | 4258.757 | 3668.452 | 3413.873 | 2842.041 | 2818.431 |
| ## 6 | 5873.427 | 5388.537 | 4906.322 | 4201.384 | 3908.009 | 3508.810 | 3069.931 | 2626.065 |
| ##   | 102      | 105.5    | 109      | 113      | 117.5    | 122      | 126.5    | 131      |
| ## 1 | 2538.021 | 2247.618 | 2274.669 | 1937.595 | 1853.359 | 1793.009 | 1756.517 | 1516.367 |
| ## 2 | 2465.159 | 2274.896 | 2123.226 | 2074.377 | 1854.472 | 1834.397 | 1711.330 | 1537.152 |
| ## 3 | 2554.263 | 2208.979 | 2153.753 | 2129.170 | 2077.445 | 1777.651 | 1621.750 | 1664.572 |
| ## 4 | 2421.829 | 2391.145 | 2377.817 | 2051.115 | 2080.636 | 1810.110 | 1731.530 | 1682.714 |
| ## 5 | 2466.716 | 2399.010 | 2210.243 | 2042.621 | 2006.983 | 1916.026 | 1796.339 | 1709.397 |
| ## 6 | 2698.418 | 2556.055 | 2142.996 | 2162.599 | 2093.221 | 1889.757 | 1784.344 | 1628.241 |

```

##      135.5      140.5      146      151.5      157      162.5      168.5      175
## 1 1402.989 1393.854 1201.337 1232.041 1038.491 842.605 966.673 986.092
## 2 1600.321 1296.363 1412.774 1154.705 1301.814 1052.601 898.659 931.269
## 3 1698.079 1398.789 1364.831 1237.431 1172.154 977.313 1076.738 959.608
## 4 1493.433 1420.170 1301.576 1294.773 1008.620 980.325 919.513 884.260
## 5 1602.446 1392.331 1289.403 1233.346 1106.326 1178.849 859.516 979.881
## 6 1557.199 1451.036 1401.997 1243.497 1236.863 1098.424 945.006 943.942
##      181      187.5      194.5      201.5      209      217      225      233      241.5
## 1 857.364 883.544 765.581 729.769 715.337 639.000 553.750 524.392 506.776
## 2 780.790 739.892 708.611 613.622 645.240 647.817 573.260 570.102 456.216
## 3 825.174 794.066 722.401 691.995 677.901 616.601 589.802 455.427 422.744
## 4 789.248 792.965 712.277 696.179 617.265 596.020 530.092 581.154 524.253
## 5 832.008 773.040 660.379 805.312 632.536 592.711 585.909 544.394 443.887
## 6 899.463 770.290 748.299 757.026 702.633 707.764 621.278 612.352 564.387
##      250.5      259.5      269      279      289      299.5      310.5      322      334
## 1 463.636 396.987 368.449 291.418 276.257 233.573 246.820 186.964 169.124
## 2 410.224 427.664 415.029 288.641 261.489 219.244 191.458 201.809 138.921
## 3 449.623 450.036 341.147 335.495 283.009 252.993 260.690 233.852 174.338
## 4 460.110 413.281 387.842 331.133 320.346 260.985 273.399 238.332 207.620
## 5 493.967 376.266 346.639 313.250 269.700 270.287 252.098 211.557 157.856
## 6 407.486 389.319 419.432 311.440 263.083 266.570 234.922 254.118 248.421
##      346      358.5      372      385.5      399.5      414.5      429.5      445      461.5      478.5
## 1 160.291 101.170 105.324 107.734 68.150 79.852 60.160 39.428 51.185 26.116
## 2 159.808 152.503 99.196 84.170 54.522 69.440 38.929 35.846 32.906 29.848
## 3 163.470 101.103 106.541 111.226 85.528 45.314 59.347 40.653 51.390 41.204
## 4 144.812 147.345 116.648 64.477 48.086 54.477 51.485 37.128 61.633 18.804
## 5 153.238 130.405 128.471 67.953 84.560 72.909 49.544 32.259 32.905 41.040
## 6 158.702 135.787 93.337 81.461 82.450 59.505 46.381 43.364 44.232 30.091
##      496      514
## 1 49.166 11.587
## 2 26.475 34.763
## 3 18.986 27.145
## 4 38.122 19.466
## 5 34.038 11.587
## 6 41.942 23.364

```